

#1 · Pharmacology

Define bioavailability.

Answer

The fraction of an administered dose of unchanged drug that reaches the systemic circulation. Intravenous drugs have a bioavailability of 100%.

#2 · Pharmacology

What is the volume of distribution (Vd)?

Answer

A theoretical volume that relates the total amount of drug in the body to the plasma concentration. A high Vd suggests extensive tissue distribution, while a low Vd suggests the drug stays mainly in plasma.

#3 · Pharmacology

Why does thiopental have a large volume of distribution despite being highly protein bound?

Answer

Thiopental is highly lipid soluble, allowing rapid redistribution into fat and muscle, which produces a large apparent V_d despite about 80% plasma protein binding.

#4 · Pharmacology

How does hypoalbuminaemia affect a highly protein-bound drug such as phenytoin?

Answer

Reduced albumin increases the free (unbound) fraction of drug, raising the pharmacologically active concentration and the risk of toxicity even when total plasma levels appear normal.

#5 · Pharmacology

What is a phase I metabolic reaction?

Answer

Oxidation, reduction or hydrolysis, usually via cytochrome P450 enzymes, that introduces or exposes a functional group, often producing an active or toxic metabolite.

#6 · Pharmacology

What is a phase II metabolic reaction?

Answer

Conjugation of a drug or its phase I metabolite with a polar group such as glucuronide, sulphate or glycine, producing an inactive, water-soluble compound for renal excretion.

#7 · Pharmacology

Define clearance.

Answer

The volume of plasma from which a drug is completely removed per unit time, expressed in mL/min or L/hour, reflecting the combined efficiency of all elimination organs.

#8 · Pharmacology

How is elimination half-life related to clearance and volume of distribution?

Answer

Half-life is proportional to V_d and inversely proportional to clearance, expressed as $t_{1/2} = 0.693 \times V_d / \text{clearance}$.

#9 · Pharmacology

Define context-sensitive half-time.

Answer

The time for the plasma concentration of a drug to fall by 50% after stopping an infusion, which depends on the duration of the infusion because of accumulation in peripheral compartments.

#10 · Pharmacology

Why does remifentanil have a short, infusion-duration-independent context-sensitive half-time?

Answer

Remifentanil is metabolised by non-specific plasma and tissue esterases rather than hepatic or renal pathways, giving rapid, predictable offset regardless of infusion length.

#11 · Pharmacology

What is zero-order kinetics?

Answer

Elimination of a constant amount of drug per unit time, independent of plasma concentration, seen with ethanol, phenytoin at high dose and aspirin at high dose.

#12 · Pharmacology

What is first-order kinetics?

Answer

Elimination of a constant proportion of the drug per unit time, so the rate of elimination is directly proportional to plasma concentration. Most anaesthetic drugs follow this pattern.

#13 · Pharmacology

What is meant by drug potency?

Answer

The dose of a drug required to produce a defined effect, commonly expressed as the ED50. A more potent drug achieves the same effect at a lower dose.

#14 · Pharmacology

What is meant by drug efficacy?

Answer

The maximum effect a drug can produce regardless of dose. A partial agonist has lower efficacy than a full agonist even at maximal receptor occupancy.

#15 · Pharmacology

Define the therapeutic index.

Answer

The ratio of the dose causing toxicity (TD50) to the dose producing the desired therapeutic effect (ED50). A high therapeutic index indicates a wider margin of safety.

#16 · Pharmacology

What is a competitive antagonist?

Answer

A drug that binds reversibly to the same receptor site as the agonist, shifting the dose-response curve to the right without reducing the maximal effect, since the block can be overcome by increasing agonist concentration.

#17 · Pharmacology

What is a non-competitive antagonist?

Answer

A drug that binds to a different site or irreversibly to the receptor, reducing the maximum achievable effect regardless of how much agonist is added.

#18 · Pharmacology

What does EC₅₀ represent on a dose-response curve?

Answer

The concentration of a drug that produces 50% of its own maximal effect, used as a measure of potency.

#19 · Pharmacology

What are the three compartments of the Marsh and Schnider TCI models for propofol?

Answer

A central compartment representing plasma, a rapidly equilibrating peripheral compartment, and a slowly equilibrating peripheral compartment, linked by inter-compartmental rate constants.

#20 · Pharmacology

How does the Schnider model differ from the Marsh model for propofol TCI?

Answer

The Schnider model incorporates age, height, weight and lean body mass into its pharmacokinetic parameters and targets effect-site concentration directly, whereas the Marsh model uses fixed compartment volumes based on weight alone and traditionally targets plasma concentration.

#21 · Pharmacology

What is the effect-site concentration in TCI?

Answer

The modelled drug concentration at the biophase (site of action, such as the brain), which lags behind plasma concentration and better predicts clinical effect than plasma concentration alone.

#22 · Pharmacology

What is ke_0 in TCI modelling?

Answer

The rate constant describing equilibration between plasma and effect-site concentrations; a higher ke_0 means faster equilibration and a shorter time to peak effect.

#23 · Pharmacology

Why do neonates have altered drug pharmacokinetics compared with adults?

Answer

Neonates have a higher proportion of total body water, immature hepatic enzyme systems, reduced plasma protein binding and a lower glomerular filtration rate, prolonging drug action and altering dosing requirements.

#24 · Pharmacology

How does ageing affect drug pharmacokinetics?

Answer

Elderly patients typically have reduced lean body mass, increased fat proportion, decreased hepatic blood flow and enzyme activity, and reduced renal clearance, prolonging drug half-lives and increasing sensitivity.

#25 · Pharmacology

How does chronic liver disease alter drug handling?

Answer

Reduced hepatic blood flow, decreased enzyme activity and low albumin production reduce the metabolism and protein binding of many drugs, increasing free drug concentrations and prolonging effect.

#26 · Pharmacology

How does renal failure affect neuromuscular blocking drug choice?

Answer

Atracurium and cisatracurium are preferred in renal failure because they are eliminated by Hofmann elimination and ester hydrolysis rather than renal excretion, unlike vecuronium and rocuronium which depend partly on renal clearance.

#27 · Pharmacology

How does pregnancy alter maternal pharmacokinetics?

Answer

Increased plasma volume and cardiac output, reduced albumin concentration, increased minute ventilation and increased hepatic and renal clearance combine to lower plasma drug concentrations for a given dose.

#28 · Pharmacology

What is the induction dose of propofol in a healthy adult?

Answer

Approximately 1.5 to 2.5 mg/kg intravenously, titrated to effect, with lower doses required in the elderly and higher doses sometimes needed in children.

#29 · Pharmacology

What is the mechanism of action of propofol?

Answer

Propofol potentiates GABA-A receptor-mediated chloride influx, increasing inhibitory neurotransmission and producing hypnosis.

#30 · Pharmacology

What cardiovascular effect is characteristic of propofol induction?

Answer

A significant fall in systemic vascular resistance and myocardial contractility causing hypotension, with attenuation of the baroreceptor reflex so heart rate often does not compensate.

#31 · Pharmacology

What is propofol infusion syndrome?

Answer

A rare but potentially fatal complication of prolonged high-dose propofol infusion characterised by metabolic acidosis, rhabdomyolysis, hyperkalaemia, cardiac failure and renal failure, most associated with doses above 4 mg/kg/hour for more than 48 hours.

#32 · Pharmacology

What is the induction dose of thiopental?

Answer

Approximately 3 to 5 mg/kg intravenously, producing loss of consciousness within one arm-brain circulation time (about 20 to 30 seconds).

#33 · Pharmacology

Why is thiopental contraindicated in porphyria?

Answer

Barbiturates induce delta-aminolaevulinic acid synthase, increasing porphyrin precursor synthesis and precipitating an acute porphyric crisis in susceptible patients.

#34 · Pharmacology

What is the induction dose and mechanism of ketamine?

Answer

1 to 2 mg/kg intravenously (or 4 to 10 mg/kg intramuscularly), acting as a non-competitive antagonist at the NMDA receptor to produce dissociative anaesthesia with preserved airway reflexes.

#35 · Pharmacology

Why is ketamine useful in a haemodynamically unstable or shocked patient?

Answer

Ketamine stimulates the sympathetic nervous system, increasing heart rate, blood pressure and cardiac output, unlike propofol and thiopental which cause cardiovascular depression.

#36 · Pharmacology

What is the induction dose of etomidate and its main advantage?

Answer

0.3 mg/kg intravenously; it produces minimal cardiovascular depression, making it useful for induction in patients with poor cardiac reserve.

#37 · Pharmacology

What is the major concern with etomidate use, particularly by infusion?

Answer

Etomidate inhibits 11-beta-hydroxylase, suppressing adrenocortical steroid synthesis and causing adrenal insufficiency, which limits its use to single induction doses rather than infusion.

#38 · Pharmacology

Define minimum alveolar concentration (MAC).

Answer

The alveolar concentration of an inhalational anaesthetic at one atmosphere that prevents a motor response to a standard surgical stimulus in 50% of subjects.

#39 · Pharmacology

How does age affect MAC?

Answer

MAC is highest in infants around six months of age and progressively decreases with increasing age, falling by approximately 6% per decade after early adulthood.

#40 · Pharmacology

List three factors that decrease MAC.

Answer

Hypothermia, pregnancy, and concurrent opioid or sedative administration all decrease MAC; acute alcohol intoxication also decreases it.

#41 · Pharmacology

What determines the speed of onset of an inhalational agent?

Answer

A low blood-gas partition coefficient allows faster equilibration between alveolar and blood partial pressures, producing a more rapid onset of anaesthesia, as seen with desflurane and sevoflurane compared with isoflurane.

#42 · Pharmacology

What is the mechanism of action of suxamethonium?

Answer

A depolarising neuromuscular blocker that binds nicotinic acetylcholine receptors at the neuromuscular junction, causing sustained depolarisation and initial fasciculation followed by flaccid paralysis.

#43 · Pharmacology

What is the intubating dose of suxamethonium and its onset time?

Answer

1 to 1.5 mg/kg intravenously, with onset of profound paralysis within 30 to 60 seconds, making it the drug of choice for rapid sequence induction.

#44 · Pharmacology

Name three contraindications to suxamethonium.

Answer

Hyperkalaemia, history of malignant hyperthermia, and major burns or neuromuscular disease beyond 24 hours (risk of exaggerated hyperkalaemic response) are all contraindications.

#45 · Pharmacology

What causes prolonged block with suxamethonium?

Answer

Reduced or atypical plasma cholinesterase activity, either from genetic variants (such as homozygous atypical alleles) or acquired causes like liver disease and pregnancy, delays hydrolysis and prolongs paralysis.

#46 · Pharmacology

What is the intubating dose and onset of rocuronium?

Answer

0.6 mg/kg intravenously gives intubating conditions in about 60 to 90 seconds; a higher dose of 1.2 mg/kg can achieve suxamethonium-like onset for rapid sequence induction when suxamethonium is contraindicated.

#47 · Pharmacology

How is atracurium eliminated?

Answer

Predominantly by Hofmann elimination (spontaneous degradation at body temperature and pH) and ester hydrolysis, making it organ-independent and suitable for hepatic or renal impairment.

#48 · Pharmacology

What metabolite of atracurium is associated with a theoretical lowering of seizure threshold?

Answer

Laudanosine, produced by Hofmann elimination, can accumulate with prolonged infusion and has been associated with CNS excitation and a theoretical lowering of seizure threshold. Histamine release, in contrast, is caused by atracurium itself, not by laudanosine.

#49 · Pharmacology

Why is cisatracurium often preferred over atracurium?

Answer

Cisatracurium is a single stereoisomer of atracurium with greater potency, fewer histamine-related side effects and less laudanosine production, while retaining organ-independent Hofmann elimination.

#50 · Pharmacology

What is the mechanism of action of neostigmine for reversal of neuromuscular block?

Answer

Neostigmine inhibits acetylcholinesterase, increasing acetylcholine concentration at the neuromuscular junction to competitively displace residual non-depolarising blocker from nicotinic receptors.

#51 · Pharmacology

Why is glycopyrrolate co-administered with neostigmine?

Answer

Neostigmine increases acetylcholine at muscarinic receptors systemically, causing bradycardia, salivation and bronchospasm; glycopyrrolate, an antimuscarinic, prevents these unwanted effects without significantly crossing the blood-brain barrier.

#52 · Pharmacology

What is the mechanism of action of sugammadex?

Answer

Sugammadex is a modified gamma-cyclodextrin that encapsulates rocuronium (and to a lesser extent vecuronium) molecules in a tight complex, rapidly reducing free plasma concentration and reversing neuromuscular block.

#53 · Pharmacology

What dose of sugammadex is used to reverse deep block for immediate reversal?

Answer

16 mg/kg intravenously is used to reverse profound rocuronium-induced block immediately after intubation, such as in a failed rapid sequence induction.

#54 · Pharmacology

What is the maximum safe dose of plain lidocaine and with adrenaline?

Answer

3 mg/kg for plain lidocaine and up to 7 mg/kg when combined with adrenaline, which reduces systemic absorption by causing local vasoconstriction.

#55 · Pharmacology

What is the first-line treatment for local anaesthetic systemic toxicity (LAST)?

Answer

Intravenous 20% lipid emulsion, given as an initial bolus of 1.5 mL/kg over one minute followed by an infusion of 15 mL/kg/hour, alongside standard resuscitation and seizure management.

#56 · Pharmacology

What is the primary G-protein and second messenger effect of alpha-1 receptor activation?

Answer

Alpha-1 receptors couple to Gq protein, activating phospholipase C to raise intracellular IP3 and diacylglycerol, which increases intracellular calcium and causes smooth muscle contraction, notably vasoconstriction.

#57 · Pharmacology

Where are alpha-2 receptors located and what is their main autonomic role?

Answer

Alpha-2 receptors are mainly presynaptic and act via Gi protein to inhibit adenylate cyclase, reducing noradrenaline release through negative feedback; central alpha-2 stimulation also produces sedation and analgesia.

#58 · Pharmacology

What second messenger pathway do beta-1 receptors use in the heart?

Answer

Beta-1 receptors couple to Gs protein, activating adenylate cyclase to increase cyclic AMP, which raises intracellular calcium and produces positive chronotropy, inotropy, dromotropy and lusitropy.

#59 · Pharmacology

What is the clinical effect of beta-2 receptor stimulation in bronchial smooth muscle?

Answer

Beta-2 stimulation raises cyclic AMP via Gs protein, causing smooth muscle relaxation, which produces bronchodilation, vasodilation in skeletal muscle beds, and uterine relaxation.

#60 · Pharmacology

What type of receptor is the muscarinic receptor and where is M3 clinically important?

Answer

Muscarinic receptors are G-protein coupled receptors; M3 receptors are found in bronchial and gut smooth muscle, salivary glands and the bladder, and mediate bronchoconstriction, increased secretions and detrusor contraction.

#61 · Pharmacology

What type of receptor is the nicotinic receptor at the neuromuscular junction?

Answer

The neuromuscular nicotinic receptor (Nm) is a ligand-gated ion channel; acetylcholine binding opens a cation channel permeable to sodium and potassium, causing end-plate depolarisation.

#62 · Pharmacology

Adrenaline at low infusion doses versus high infusion doses: what receptor shift occurs?

Answer

At low doses (around 0.01 to 0.05 micrograms/kg/min) beta effects predominate causing vasodilation and increased contractility; at higher doses alpha-1 vasoconstriction dominates, raising systemic vascular resistance.

#63 · Pharmacology

What is the standard adult intravenous bolus dose of adrenaline for cardiac arrest?

Answer

1 mg (10 mL of 1 in 10,000) intravenously every 3 to 5 minutes during adult cardiac arrest, given after the third shock in a shockable rhythm and as soon as possible in a non-shockable rhythm.

#64 · Pharmacology

What are the main receptor actions of noradrenaline used as an infusion?

Answer

Noradrenaline is a potent alpha-1 agonist with modest beta-1 activity, producing marked vasoconstriction and a rise in systemic vascular resistance with relatively little change in heart rate, making it the first-line vasopressor in septic shock.

#65 · Pharmacology

What is metaraminol and what is its typical intravenous bolus dose intraoperatively?

Answer

Metaraminol is a predominantly alpha-1 agonist vasopressor (with some indirect noradrenaline release) used to treat intraoperative hypotension, typically given as 0.5 to 1 mg intravenous boluses titrated to effect.

#66 · Pharmacology

How does ephedrine raise blood pressure and why does it show tachyphylaxis?

Answer

Ephedrine is an indirect and direct sympathomimetic that stimulates release of noradrenaline from presynaptic vesicles while also directly stimulating alpha and beta receptors; tachyphylaxis occurs because repeated doses deplete presynaptic noradrenaline stores.

#67 · Pharmacology

What is the typical intravenous bolus dose of ephedrine and its usual duration of action?

Answer

Ephedrine is given as 3 to 6 mg intravenous boluses, with onset within a minute and a duration of action of around 10 to 60 minutes.

#68 · Pharmacology

What is phenylephrine's receptor selectivity and its main haemodynamic effect?

Answer

Phenylephrine is a pure alpha-1 agonist with negligible beta activity, producing vasoconstriction and a rise in blood pressure often accompanied by reflex bradycardia.

#69 · Pharmacology

Why is phenylephrine often preferred over ephedrine for hypotension during spinal anaesthesia for caesarean section?

Answer

Phenylephrine causes less fetal acidosis than ephedrine because it does not cross the placenta to the same extent and does not stimulate fetal beta-adrenergic metabolism, so it is now the preferred first-line vasopressor in obstetric spinal hypotension.

#70 · Pharmacology

What receptor does vasopressin act on to cause vasoconstriction, and what is its use in septic shock?

Answer

Vasopressin acts on V1 receptors on vascular smooth muscle via a phospholipase C pathway causing vasoconstriction independent of adrenergic receptors; it is used as a second-line agent added to noradrenaline in refractory septic shock, typically at fixed low dose infusion around 0.03 units per minute.

#71 · Pharmacology

What is the mechanism of action of dobutamine?

Answer

Dobutamine is a synthetic catecholamine that predominantly stimulates beta-1 receptors, increasing myocardial contractility and cardiac output with relatively little effect on heart rate at low doses, and mild beta-2 mediated vasodilation.

#72 · Pharmacology

What class of drug is milrinone and what is its mechanism?

Answer

Milrinone is a phosphodiesterase III inhibitor that prevents breakdown of cyclic AMP in cardiac and vascular smooth muscle, producing an inotropic and vasodilator (inodilator) effect independent of adrenergic receptors.

#73 · Pharmacology

Why should milrinone be used cautiously in a patient with poor renal function?

Answer

Milrinone is renally excreted, so its half-life is prolonged in renal impairment, increasing the risk of prolonged hypotension and arrhythmia; dose reduction is required.

#74 · Pharmacology

What is the mechanism of action of beta-blockers such as bisoprolol used for perioperative rate control?

Answer

Beta-blockers competitively antagonise beta-1 adrenoceptors, reducing heart rate, contractility and myocardial oxygen demand, and reducing renin release from the juxtaglomerular apparatus.

#75 · Pharmacology

How do dihydropyridine calcium channel blockers such as amlodipine lower blood pressure?

Answer

Dihydropyridines block L-type calcium channels predominantly in vascular smooth muscle, reducing calcium influx and causing arteriolar vasodilation with relatively little direct effect on cardiac conduction.

#76 · Pharmacology

How does verapamil differ from dihydropyridine calcium channel blockers in its cardiac effects?

Answer

Verapamil has greater affinity for cardiac L-type calcium channels than vascular ones, causing significant negative chronotropy, dromotropy and inotropy, and should not be combined with beta-blockers due to the risk of severe bradycardia and heart block.

#77 · Pharmacology

What are the main mechanisms of action of amiodarone?

Answer

Amiodarone is a class III antiarrhythmic that blocks potassium channels to prolong the action potential and refractory period, but also has class I sodium channel, class II beta-blocking and class IV calcium channel blocking properties.

#78 · Pharmacology

What is the dose of intravenous amiodarone used in adult shockable cardiac arrest?

Answer

300 mg intravenous amiodarone is given after the third shock in shockable cardiac arrest, with a further 150 mg dose available after the fifth shock if the rhythm remains shockable.

#79 · Pharmacology

Give two important adverse effects of long-term amiodarone therapy relevant to anaesthesia.

Answer

Long-term amiodarone use can cause thyroid dysfunction (both hypo- and hyperthyroidism due to its iodine content) and pulmonary fibrosis, both of which are relevant to preoperative assessment.

#80 · Pharmacology

What is the mechanism of action of adenosine and its typical use?

Answer

Adenosine stimulates A1 receptors in the atrioventricular node, activating potassium channels and causing transient hyperpolarisation and slowed conduction, briefly blocking the atrioventricular node; it is used to terminate supraventricular tachycardia involving the atrioventricular node.

#81 · Pharmacology

What is the standard dosing regimen for intravenous adenosine in adult SVT?

Answer

An initial rapid intravenous bolus of 6 mg followed by a fast saline flush, escalating to 12 mg and then a further 12 mg if the rhythm does not revert, each given through a large proximal cannula.

#82 · Pharmacology

Why must patients be warned before giving adenosine, and what condition is it contraindicated in?

Answer

Patients should be warned of a transient sensation of impending doom, flushing and chest tightness; adenosine is contraindicated in asthma because it can precipitate bronchospasm, and in second or third degree heart block without a pacemaker.

#83 · Pharmacology

What is the mechanism of action of atropine?

Answer

Atropine is a competitive antagonist at muscarinic acetylcholine receptors, blocking parasympathetic (vagal) tone and thereby increasing heart rate, reducing secretions and causing pupillary dilation.

#84 · Pharmacology

What is the typical intravenous dose of atropine used for symptomatic bradycardia?

Answer

500 micrograms intravenously, repeated up to a maximum of 3 mg, in line with Resuscitation Council UK adult bradycardia guidance.

#85 · Pharmacology

Why does glycopyrrolate cause less central nervous system disturbance than atropine or hyoscine?

Answer

Glycopyrrolate is a quaternary ammonium compound and does not cross the blood-brain barrier, so unlike atropine and hyoscine it does not cause central anticholinergic effects such as confusion or sedation.

#86 · Pharmacology

Why is glycopyrrolate combined with neostigmine for reversal of neuromuscular block?

Answer

Neostigmine increases acetylcholine at both nicotinic and muscarinic receptors, so it is combined with the antimuscarinic glycopyrrolate to prevent muscarinic side effects such as bradycardia, bronchorrhoea and salivation while allowing reversal of neuromuscular block.

#87 · Pharmacology

What are the clinical uses of hyoscine (scopolamine) in anaesthetic practice?

Answer

Hyoscine is an antimuscarinic used as an antisialagogue and, via a transdermal patch, for prevention of postoperative nausea and vomiting; it readily crosses the blood-brain barrier and can cause sedation or central anticholinergic syndrome, particularly in the elderly.

#88 · Pharmacology

What is the mechanism of action of salbutamol as a bronchodilator?

Answer

Salbutamol is a short-acting selective beta-2 agonist that increases cyclic AMP in bronchial smooth muscle causing relaxation and bronchodilation; nebulised or intravenous salbutamol is also used to treat hyperkalaemia by driving potassium intracellularly.

#89 · Pharmacology

What is the mechanism of action of ipratropium bromide?

Answer

Ipratropium bromide is a short-acting muscarinic antagonist that blocks M3 receptors in bronchial smooth muscle, reducing bronchoconstriction; it is often nebulised in combination with salbutamol for acute severe asthma or COPD exacerbations.

#90 · Pharmacology

Why is perioperative hydrocortisone supplementation given to patients on long-term steroids?

Answer

Long-term exogenous steroid use suppresses the hypothalamic-pituitary-adrenal axis, so patients cannot mount an adequate endogenous cortisol response to surgical stress; supplemental hydrocortisone (commonly 50 to 100 mg intravenously at induction, with further doses depending on surgical severity) prevents an Addisonian crisis.

#91 · Pharmacology

What is the mechanism of action of ondansetron as an antiemetic?

Answer

Ondansetron is a selective 5-HT₃ receptor antagonist that blocks serotonin receptors both centrally in the chemoreceptor trigger zone and peripherally on vagal afferents in the gut, reducing nausea and vomiting.

#92 · Pharmacology

Give an important cardiac side effect of ondansetron relevant to preoperative assessment.

Answer

Ondansetron can prolong the QT interval, so it should be used cautiously or avoided in patients with congenital long QT syndrome or on other QT-prolonging medications.

#93 · Pharmacology

What is the proposed mechanism by which dexamethasone reduces postoperative nausea and vomiting?

Answer

The exact mechanism is not fully established but dexamethasone is thought to reduce prostaglandin synthesis and central serotonin release, and its anti-inflammatory and antiemetic effect is typically given as an 4 to 8 mg intravenous dose at induction.

#94 · Pharmacology

What is the mechanism of action of cyclizine as an antiemetic?

Answer

Cyclizine is an antihistamine (H1 receptor antagonist) with anticholinergic properties acting mainly on the vomiting centre and vestibular pathways, making it particularly useful for motion-related and opioid-induced nausea.

#95 · Pharmacology

Give a notable cardiovascular side effect of intravenous cyclizine.

Answer

Rapid intravenous cyclizine can cause tachycardia and, particularly in patients with cardiac disease, may precipitate hypotension or exacerbate heart failure due to its anticholinergic and mild negative inotropic effects.

#96 · Pharmacology

What is the mechanism of action of droperidol as an antiemetic and its main safety concern?

Answer

Droperidol is a butyrophenone dopamine D2 receptor antagonist acting at the chemoreceptor trigger zone; its main safety concern is dose-dependent QT prolongation, which led to restricted licensed doses (typically 0.625 to 1.25 mg) and ECG monitoring recommendations.

#97 · Pharmacology

What is the mechanism of action of unfractionated heparin?

Answer

Unfractionated heparin binds to antithrombin III and potentiates its inhibition of thrombin (factor IIa) and factor Xa, producing an immediate anticoagulant effect that is monitored using the activated partial thromboplastin time (APTT).

#98 · Pharmacology

How does low molecular weight heparin differ in mechanism from unfractionated heparin?

Answer

Low molecular weight heparin preferentially potentiates antithrombin III's inhibition of factor Xa with much less effect on thrombin, giving a more predictable dose response that does not usually require routine monitoring.

#99 · Pharmacology

What is the minimum recommended interval between a prophylactic dose of low molecular weight heparin and neuraxial block insertion or removal?

Answer

At least 12 hours after a prophylactic dose of low molecular weight heparin before neuraxial block or catheter removal, and the next dose should be delayed at least 4 hours after the procedure, per Association of Anaesthetists guidance.

#100 · Pharmacology

What is the minimum recommended interval between a treatment (therapeutic) dose of low molecular weight heparin and neuraxial block?

Answer

At least 24 hours should elapse after a treatment dose of low molecular weight heparin before neuraxial block insertion or catheter removal.

#101 · Pharmacology

What is the mechanism of action of warfarin?

Answer

Warfarin inhibits vitamin K epoxide reductase, preventing the recycling of vitamin K needed to gamma-carboxylate and activate clotting factors II, VII, IX and X, and proteins C and S.

#102 · Pharmacology

How is warfarin's anticoagulant effect monitored and reversed in an emergency?

Answer

Warfarin is monitored using the international normalised ratio (INR); emergency reversal for major bleeding uses intravenous vitamin K plus prothrombin complex concentrate for rapid factor replacement.

#103 · Pharmacology

What is the mechanism of action of dabigatran and how is its effect reversed?

Answer

Dabigatran is a direct thrombin (factor IIa) inhibitor; its anticoagulant effect can be rapidly reversed using idarucizumab, a monoclonal antibody fragment that binds dabigatran directly.

#104 · Pharmacology

What is the mechanism of action of apixaban and rivaroxaban?

Answer

Apixaban and rivaroxaban are direct oral factor Xa inhibitors that block the conversion of prothrombin to thrombin by inhibiting free and clot-bound factor Xa.

#105 · Pharmacology

What agent can be used to reverse factor Xa inhibitors before urgent surgery or neuraxial block?

Answer

Andexanet alfa is a recombinant modified factor Xa decoy protein licensed to reverse apixaban and rivaroxaban in life-threatening bleeding; prothrombin complex concentrate is used where andexanet alfa is unavailable.

#106 · Pharmacology

What is the recommended interval between the last dose of a direct oral anticoagulant and neuraxial block, in a patient with normal renal function?

Answer

Typically at least 72 hours since the last dose for standard direct oral anticoagulants used at treatment doses in a patient with normal renal function, though the exact interval depends on the specific agent and renal function, per Association of Anaesthetists guidance.

#107 · Pharmacology

What is the mechanism of action and typical clinical use of aspirin regarding neuraxial anaesthesia timing?

Answer

Aspirin irreversibly inhibits cyclo-oxygenase-1, blocking thromboxane A2 production and platelet aggregation for the lifespan of the platelet; low-dose aspirin alone is not considered a contraindication to neuraxial block and does not require cessation.

#108 · Pharmacology

What is the mechanism of action of clopidogrel and the recommended interval before neuraxial block?

Answer

Clopidogrel irreversibly inhibits the platelet P2Y12 receptor, blocking ADP-mediated platelet aggregation; it should generally be stopped for 7 days before neuraxial block to allow platelet turnover.

#109 · Pharmacology

Which vasopressor receptor mechanism explains why noradrenaline causes reflex bradycardia at higher doses?

Answer

Marked alpha-1 mediated vasoconstriction raises systemic vascular resistance and blood pressure, triggering the baroreceptor reflex, which increases vagal tone and can produce a compensatory reflex bradycardia despite noradrenaline's mild direct beta-1 activity.

#110 · Pharmacology

Why is dobutamine sometimes preferred over adrenaline in cardiogenic shock with preserved blood pressure?

Answer

Dobutamine increases contractility and cardiac output predominantly through beta-1 stimulation with only mild vasodilatory beta-2 effect, avoiding the marked increase in afterload and myocardial oxygen demand that adrenaline's alpha-mediated vasoconstriction can cause.

#111 · Physiology Anatomy and Biochemistry

What is tidal volume (TV)?

Answer

The volume of air moved in or out of the lungs during a single normal breath, approximately 500 mL in a healthy adult at rest.

#112 · Physiology Anatomy and Biochemistry

Define inspiratory reserve volume (IRV).

Answer

The additional volume of air that can be forcibly inhaled after a normal tidal inspiration, approximately 2500 to 3000 mL.

#113 · Physiology Anatomy and Biochemistry

Define expiratory reserve volume (ERV).

Answer

The additional volume of air that can be forcibly exhaled after a normal tidal expiration, approximately 1000 to 1200 mL.

#114 · Physiology Anatomy and Biochemistry

What is residual volume (RV) and why can it not be measured by spirometry?

Answer

Residual volume is the air remaining in the lungs after maximal forced expiration, approximately 1200 mL. It cannot be measured by spirometry because it is never exhaled, so gas dilution, body plethysmography or nitrogen washout must be used instead.

#115 · Physiology Anatomy and Biochemistry

What is vital capacity (VC)?

Answer

The maximum volume of air that can be exhaled after a maximal inspiration, equal to IRV plus TV plus ERV, approximately 4500 to 5000 mL.

#116 · Physiology Anatomy and Biochemistry

What is functional residual capacity (FRC) and what does it consist of?

Answer

FRC is the volume of air remaining in the lungs at the end of a normal tidal expiration, equal to ERV plus RV, approximately 2200 to 2400 mL. It represents the resting lung volume determined by the balance between chest wall outward recoil and lung inward recoil.

#117 · Physiology Anatomy and Biochemistry

What is total lung capacity (TLC)?

Answer

The total volume of air in the lungs after a maximal inspiration, equal to VC plus RV, approximately 6000 mL.

#118 · Physiology Anatomy and Biochemistry

Why does FRC fall under general anaesthesia?

Answer

Loss of muscle tone, particularly diaphragmatic tone, causes cephalad displacement of the diaphragm, and supine positioning reduces chest wall compliance, together reducing FRC by around 20 per cent, which promotes atelectasis and shunting.

#119 · Physiology Anatomy and Biochemistry

What is closing capacity and why is it clinically important?

Answer

Closing capacity is the lung volume at which small airways in dependent lung zones begin to close during expiration. When closing capacity exceeds FRC, airway closure occurs during normal tidal breathing, causing ventilation-perfusion mismatch, which is common in the elderly, obese and pregnant patients.

#120 · Physiology Anatomy and Biochemistry

Describe the shape and physiological significance of the oxyhaemoglobin dissociation curve.

Answer

The curve is sigmoid shaped because of cooperative binding between haemoglobin's four haem subunits. The steep portion allows large amounts of oxygen to be released for a small drop in PaO_2 at the tissues, while the flat upper portion means SaO_2 stays well preserved despite a fall in PaO_2 at altitude or with mild lung disease.

#121 · Physiology Anatomy and Biochemistry

What does the P50 of the oxyhaemoglobin dissociation curve represent?

Answer

The partial pressure of oxygen at which haemoglobin is 50 per cent saturated, normally about 3.5 kPa (26.6 mmHg). A higher P50 indicates a right shift and reduced haemoglobin oxygen affinity.

#122 · Physiology Anatomy and Biochemistry

List factors that cause a right shift of the oxyhaemoglobin dissociation curve.

Answer

Increased temperature, increased hydrogen ion concentration (acidosis), increased carbon dioxide (Bohr effect) and increased 2,3-diphosphoglycerate (2,3-DPG) all shift the curve rightward, reducing haemoglobin's affinity for oxygen and favouring offloading at the tissues.

#123 · Physiology Anatomy and Biochemistry

List factors that cause a left shift of the oxyhaemoglobin dissociation curve.

Answer

Decreased temperature, decreased hydrogen ion concentration (alkalosis), decreased carbon dioxide, decreased 2,3-DPG, carbon monoxide and fetal haemoglobin all shift the curve leftward, increasing haemoglobin's affinity for oxygen and impairing offloading at the tissues.

#124 · Physiology Anatomy and Biochemistry

What is the Bohr effect?

Answer

A rise in carbon dioxide tension or hydrogen ion concentration in tissue capillaries reduces haemoglobin's affinity for oxygen, shifting the dissociation curve to the right and promoting oxygen release where it is metabolically needed.

#125 · Physiology Anatomy and Biochemistry

What is the role of 2,3-diphosphoglycerate (2,3-DPG) in oxygen transport?

Answer

2,3-DPG is produced by red blood cells during glycolysis and binds preferentially to deoxygenated haemoglobin, stabilising the low affinity conformation and shifting the curve to the right, which promotes oxygen unloading at the tissues.

#126 · Physiology Anatomy and Biochemistry

Why does stored blood have reduced 2,3-DPG and what is the clinical consequence?

Answer

Storage in citrate anticoagulant depletes red cell 2,3-DPG, causing a left shift of the dissociation curve so transfused blood initially has an increased affinity for oxygen and releases it less readily to tissues until 2,3-DPG levels are restored, over roughly 24 hours.

#127 · Physiology Anatomy and Biochemistry

Define ventilation-perfusion (V/Q) matching.

Answer

The ratio of alveolar ventilation to pulmonary capillary blood flow within a lung unit. Ideal gas exchange requires ventilation and perfusion to be matched, with a whole-lung V/Q ratio of approximately 0.8.

#128 · Physiology Anatomy and Biochemistry

How does V/Q ratio vary from the apex to the base of the upright lung?

Answer

Both ventilation and perfusion increase from apex to base due to gravity, but perfusion increases proportionally more, so the V/Q ratio is highest (ventilation exceeds perfusion) at the apex and lowest (perfusion exceeds ventilation) at the base.

#129 · Physiology Anatomy and Biochemistry

What is the physiological consequence of a V/Q ratio of zero versus infinity?

Answer

A V/Q ratio of zero represents shunt, where blood perfuses unventilated alveoli and gas exchange cannot occur. A V/Q ratio of infinity represents dead space, where alveoli are ventilated but not perfused, so no gas exchange occurs there either.

#130 · Physiology Anatomy and Biochemistry

What is hypoxic pulmonary vasoconstriction (HPV)?

Answer

A local reflex in which pulmonary arterioles supplying poorly ventilated alveoli constrict in response to low alveolar oxygen tension, diverting blood flow towards better-ventilated lung units and thereby minimising V/Q mismatch.

#131 · Physiology Anatomy and Biochemistry

How do volatile anaesthetic agents affect hypoxic pulmonary vasoconstriction?

Answer

Volatile anaesthetic agents attenuate hypoxic pulmonary vasoconstriction in a dose-dependent manner, which can worsen V/Q mismatch and reduce arterial oxygenation, particularly relevant during one-lung ventilation.

#132 · Physiology Anatomy and Biochemistry

Where are the central chemoreceptors located and what do they respond to?

Answer

Central chemoreceptors lie on the ventral surface of the medulla oblongata and respond to changes in the hydrogen ion concentration of the cerebrospinal fluid, which reflects arterial carbon dioxide tension since carbon dioxide crosses the blood-brain barrier freely.

#133 · Physiology Anatomy and Biochemistry

Where are the peripheral chemoreceptors located and what do they detect?

Answer

Peripheral chemoreceptors are located in the carotid bodies at the bifurcation of the common carotid arteries and in the aortic bodies. They detect falls in arterial oxygen tension, and to a lesser extent rises in carbon dioxide tension and hydrogen ion concentration.

#134 · Physiology Anatomy and Biochemistry

Which chemoreceptors provide the main drive to ventilation under normal conditions?

Answer

The central chemoreceptors provide the dominant drive to ventilation, accounting for approximately 80 per cent of the response to a rise in arterial carbon dioxide tension, with the peripheral chemoreceptors contributing the remainder.

#135 · Physiology Anatomy and Biochemistry

What is the hypoxic ventilatory response and how is it affected by anaesthesia?

Answer

The hypoxic ventilatory response is the increase in ventilation driven by peripheral chemoreceptors when arterial oxygen tension falls, typically below about 8 kPa. Volatile anaesthetic agents and opioids markedly blunt this response even at subanaesthetic concentrations.

#136 · Physiology Anatomy and Biochemistry

Which brainstem centres generate the basic respiratory rhythm?

Answer

The dorsal respiratory group and ventral respiratory group in the medulla generate the basic rhythm of breathing, with the pontine respiratory group (pneumotaxic centre) fine-tuning the rate and depth of breathing.

#137 · Physiology Anatomy and Biochemistry

What is the Hering-Breuer inflation reflex?

Answer

Stretch receptors in the airway smooth muscle, activated by lung inflation, send inhibitory signals via the vagus nerve to terminate inspiration and prevent overinflation of the lungs.

#138 · Physiology Anatomy and Biochemistry

Outline the phases of the cardiac cycle in order.

Answer

Atrial systole, isovolumetric ventricular contraction, ventricular ejection, isovolumetric ventricular relaxation, and ventricular filling, which itself has rapid and slow (diastasis) phases before the cycle repeats with the next atrial systole.

#139 · Physiology Anatomy and Biochemistry

What causes the first heart sound (S1)?

Answer

Closure of the mitral and tricuspid (atrioventricular) valves at the start of isovolumetric ventricular contraction, marking the beginning of systole.

#140 · Physiology Anatomy and Biochemistry

What causes the second heart sound (S2)?

Answer

Closure of the aortic and pulmonary (semilunar) valves at the end of ventricular ejection, marking the beginning of diastole.

#141 · Physiology Anatomy and Biochemistry

What is isovolumetric ventricular contraction?

Answer

The phase of the cardiac cycle after the atrioventricular valves close but before ventricular pressure exceeds aortic and pulmonary pressures, during which the ventricles contract with no change in volume because all valves are closed.

#142 · Physiology Anatomy and Biochemistry

What produces the 'a wave' on the jugular venous pressure waveform?

Answer

The a wave is produced by atrial contraction (atrial systole) immediately before ventricular systole, and it is absent in atrial fibrillation.

#143 · Physiology Anatomy and Biochemistry

What produces the 'v wave' on the jugular venous pressure waveform?

Answer

The v wave reflects venous filling of the right atrium against a closed tricuspid valve during ventricular systole, peaking just before the tricuspid valve opens.

#144 · Physiology Anatomy and Biochemistry

State the equation for cardiac output and normal resting value.

Answer

Cardiac output equals stroke volume multiplied by heart rate. In a healthy adult at rest this is approximately 5 litres per minute (stroke volume around 70 mL, heart rate around 70 beats per minute).

#145 · Physiology Anatomy and Biochemistry

What are the three main determinants of stroke volume?

Answer

Preload, afterload and myocardial contractility. Heart rate is not itself a determinant of stroke volume; together with stroke volume it determines cardiac output ($CO = SV \times HR$), although very high heart rates can secondarily reduce stroke volume by shortening diastolic filling time.

#146 · Physiology Anatomy and Biochemistry

Define preload and afterload.

Answer

Preload is the degree of myocardial fibre stretch at the end of diastole, closely related to ventricular end-diastolic volume. Afterload is the resistance or tension the ventricle must overcome to eject blood, closely related to systemic (or pulmonary) vascular resistance.

#147 · Physiology Anatomy and Biochemistry

State the Frank-Starling law of the heart.

Answer

Within physiological limits, the force of ventricular contraction is proportional to the initial length of the myocardial fibres, so an increase in venous return and end-diastolic volume produces a corresponding increase in stroke volume.

#148 · Physiology Anatomy and Biochemistry

What is the physiological mechanism behind the Frank-Starling relationship?

Answer

Increased sarcomere stretch improves the overlap and calcium sensitivity of actin and myosin filaments, increasing the number of effective cross-bridges formed and thereby increasing the force of contraction, up to an optimal sarcomere length beyond which force declines.

#149 · Physiology Anatomy and Biochemistry

What happens to the Frank-Starling curve in heart failure?

Answer

The curve shifts down and to the right, meaning that for any given preload, stroke volume is reduced compared with the normal heart, reflecting impaired contractility.

#150 · Physiology Anatomy and Biochemistry

What is the sinoatrial (SA) node and why is it the natural pacemaker?

Answer

The SA node is a group of specialised pacemaker cells located in the wall of the right atrium near the opening of the superior vena cava. It has the fastest intrinsic rate of spontaneous depolarisation of all cardiac conduction tissue, so it normally sets the heart rate.

#151 · Physiology Anatomy and Biochemistry

Trace the normal cardiac conduction pathway from the SA node to the ventricles.

Answer

Impulses spread from the SA node through the atria to the atrioventricular (AV) node, then down the bundle of His, into the left and right bundle branches, and finally through the Purkinje fibres to the ventricular myocardium.

#152 · Physiology Anatomy and Biochemistry

Why does the AV node conduct impulses slowly?

Answer

The AV node has slow-conducting cells and small fibre diameter, which delays the impulse by about 0.1 second, allowing time for atrial contraction to complete ventricular filling before ventricular systole begins.

#153 · Physiology Anatomy and Biochemistry

What are the intrinsic firing rates of the SA node, AV node and Purkinje fibres?

Answer

The SA node fires at approximately 60 to 100 beats per minute, the AV node (junctional tissue) at approximately 40 to 60 beats per minute, and the Purkinje fibres (ventricular escape rhythm) at approximately 20 to 40 beats per minute.

#154 · Physiology Anatomy and Biochemistry

Describe phase 0 of the cardiac action potential in ventricular myocytes.

Answer

Phase 0 is rapid depolarisation caused by the fast opening of voltage-gated sodium channels, producing a rapid influx of sodium ions and a steep upstroke of the action potential.

#155 · Physiology Anatomy and Biochemistry

What underlies the plateau phase (phase 2) of the ventricular action potential?

Answer

Phase 2 is maintained by a balance between slow inward calcium current through L-type calcium channels and outward potassium current, and it corresponds to the myocardial refractory period, which prevents tetanic contraction.

#156 · Physiology Anatomy and Biochemistry

From which arteries does the heart receive its blood supply?

Answer

The heart is supplied by the left and right coronary arteries, which arise from the left and right aortic sinuses (sinuses of Valsalva) just above the aortic valve cusps.

#157 · Physiology Anatomy and Biochemistry

When during the cardiac cycle does left ventricular coronary blood flow mainly occur, and why?

Answer

Left ventricular coronary flow occurs predominantly during diastole, because during systole the high intramyocardial pressure generated by contraction compresses the coronary vessels and largely occludes flow, particularly in the subendocardium.

#158 · Physiology Anatomy and Biochemistry

How does myocardial oxygen extraction compare with other tissues, and what is the implication for increased demand?

Answer

The myocardium extracts approximately 60 to 70 per cent of the oxygen delivered to it at rest, far more than most other tissues. Because extraction is already near-maximal, increased myocardial oxygen demand must be met almost entirely by increasing coronary blood flow rather than by extracting more oxygen.

#159 · Physiology Anatomy and Biochemistry

What is the principal local regulator of coronary blood flow?

Answer

Coronary blood flow is regulated mainly by local metabolic autoregulation, with adenosine released from myocytes during increased metabolic activity acting as a potent coronary vasodilator that matches flow to myocardial oxygen demand.

#160 · Physiology Anatomy and Biochemistry

What is coronary perfusion pressure and how is it calculated?

Answer

Coronary perfusion pressure is the pressure gradient driving blood flow through the coronary arteries, calculated as aortic diastolic pressure minus left ventricular end-diastolic pressure.

#161 · Physiology Anatomy and Biochemistry

What is the initial compensatory response to acute moderate haemorrhage?

Answer

A fall in arterial pressure unloads the arterial baroreceptors, reducing their inhibitory input to the vasomotor centre, which increases sympathetic outflow, causing tachycardia, increased myocardial contractility and peripheral vasoconstriction to maintain arterial pressure.

#162 · Physiology Anatomy and Biochemistry

How does the body redistribute blood flow during haemorrhage?

Answer

Sympathetically mediated vasoconstriction preferentially reduces blood flow to the skin, skeletal muscle, kidneys and splanchnic circulation, preserving perfusion to the heart and brain, which have limited autoregulatory reserve and depend on maintained flow.

#163 · Physiology Anatomy and Biochemistry

Which hormonal systems are activated during haemorrhage to restore blood volume?

Answer

The renin-angiotensin-aldosterone system is activated by reduced renal perfusion, causing vasoconstriction and sodium and water retention. Antidiuretic hormone (vasopressin) release increases from reduced atrial stretch, promoting water reabsorption and vasoconstriction at higher concentrations.

#164 · Physiology Anatomy and Biochemistry

What is transcapillary refill and how does it contribute to compensation for haemorrhage?

Answer

Reduced capillary hydrostatic pressure following blood loss favours the net movement of interstitial fluid into the vascular space, partially restoring circulating volume over the following hours, though this dilutes plasma proteins and haematocrit.

#165 · Physiology Anatomy and Biochemistry

What is class III haemorrhage according to the Advanced Trauma Life Support classification?

Answer

Class III haemorrhage represents a blood loss of approximately 30 to 40 per cent of circulating volume (roughly 1500 to 2000 mL in an adult), typically causing marked tachycardia, tachypnoea, falling blood pressure and a decreased conscious level, requiring blood product transfusion.

#166 · Physiology Anatomy and Biochemistry

What is the normal glomerular filtration rate (GFR) in a healthy adult?

Answer

Approximately 100 to 125 mL/min, or about 180 L/day of filtrate produced across both kidneys.

#167 · Physiology Anatomy and Biochemistry

How is GFR calculated using clearance principles?

Answer

GFR equals the clearance of a freely filtered, non-reabsorbed, non-secreted marker such as inulin: $\text{clearance} = (\text{urine concentration} \times \text{urine flow rate}) \div \text{plasma concentration}$.

#168 · Physiology Anatomy and Biochemistry

Over what range of mean arterial pressure does renal blood flow autoregulate?

Answer

Renal blood flow and GFR remain relatively constant between a mean arterial pressure of about 80 and 180 mmHg, mainly via the myogenic reflex and tubuloglomerular feedback.

#169 · Physiology Anatomy and Biochemistry

What is tubuloglomerular feedback?

Answer

The macula densa senses sodium chloride delivery to the distal tubule; increased delivery causes afferent arteriolar constriction, reducing GFR to protect against excessive filtration.

#170 · Physiology Anatomy and Biochemistry

What percentage of cardiac output do the kidneys normally receive?

Answer

The kidneys receive approximately 20 to 25% of cardiac output despite representing only about 0.5% of body mass.

#171 · Physiology Anatomy and Biochemistry

How does positive pressure ventilation affect renal perfusion?

Answer

Positive pressure ventilation raises intrathoracic pressure, reduces venous return and cardiac output, and activates the renin-angiotensin-aldosterone system, reducing renal blood flow and urine output.

#172 · Physiology Anatomy and Biochemistry

How does general anaesthesia typically affect renal blood flow and GFR?

Answer

Most volatile and intravenous anaesthetic agents reduce renal blood flow and GFR by lowering mean arterial pressure and cardiac output and by increasing sympathetic tone and antidiuretic hormone release.

#173 · Physiology Anatomy and Biochemistry

Where is approximately 65% of filtered sodium and water reabsorbed?

Answer

The proximal convoluted tubule reabsorbs about 65% of filtered sodium and water, largely isototically alongside glucose and amino acids.

#174 · Physiology Anatomy and Biochemistry

What is the site of action and mechanism of loop diuretics?

Answer

Loop diuretics such as furosemide act on the thick ascending limb of the loop of Henle, inhibiting the Na-K-2Cl cotransporter and reducing the medullary concentration gradient.

#175 · Physiology Anatomy and Biochemistry

What hormone acts on the distal tubule and collecting duct to increase sodium reabsorption?

Answer

Aldosterone, released from the adrenal cortex in response to angiotensin II or hyperkalaemia, increases sodium reabsorption and potassium and hydrogen ion secretion.

#176 · Physiology Anatomy and Biochemistry

What is the Henderson-Hasselbalch equation for the bicarbonate buffer system?

Answer

$$\text{pH} = 6.1 + \log_{10}([\text{HCO}_3^-] / (0.03 \times \text{PaCO}_2))$$
, relating blood pH to the ratio of bicarbonate to dissolved carbon dioxide.

#177 · Physiology Anatomy and Biochemistry

How is the anion gap calculated and what is its normal range?

Answer

Anion gap = $[\text{Na}^+]$ minus $([\text{Cl}^-] + [\text{HCO}_3^-])$, normally 8 to 16 mmol/L; it represents unmeasured anions such as albumin, phosphate and sulfate.

#178 · Physiology Anatomy and Biochemistry

Give two causes of a high anion gap metabolic acidosis.

Answer

Diabetic ketoacidosis and lactic acidosis are classic causes; other causes include renal failure, salicylate poisoning, and methanol or ethylene glycol ingestion.

#179 · Physiology Anatomy and Biochemistry

Give a cause of a normal anion gap (hyperchloraemic) metabolic acidosis.

Answer

Diarrhoea causes loss of bicarbonate-rich fluid, and renal tubular acidosis and large-volume 0.9% saline administration are other common causes.

#180 · Physiology Anatomy and Biochemistry

How does the body compensate for a primary metabolic acidosis?

Answer

Respiratory compensation occurs through hyperventilation, lowering PaCO_2 ; Winter's formula predicts expected PaCO_2 as $(1.5 \times [\text{HCO}_3^-]) + 8$, plus or minus 2.

#181 · Physiology Anatomy and Biochemistry

How does the body compensate for a primary respiratory acidosis?

Answer

The kidneys retain bicarbonate over hours to days (metabolic compensation), raising serum bicarbonate to partially normalise pH; acute compensation is minimal, chronic compensation is more complete.

#182 · Physiology Anatomy and Biochemistry

What is base excess and what does a negative value indicate?

Answer

Base excess is the amount of acid or base needed to titrate one litre of blood to pH 7.40 at a PaCO_2 of 5.3 kPa; a negative base excess (base deficit) indicates a metabolic acidosis.

#183 · Physiology Anatomy and Biochemistry

What are common causes of a respiratory alkalosis?

Answer

Hyperventilation from anxiety, pain, pregnancy, hypoxaemia, mechanical overventilation, or central causes such as raised intracranial pressure or salicylate toxicity.

#184 · Physiology Anatomy and Biochemistry

What are common causes of a metabolic alkalosis?

Answer

Vomiting or nasogastric losses of hydrochloric acid, loop or thiazide diuretic use, and excess mineralocorticoid activity such as in Conn's syndrome.

#185 · Physiology Anatomy and Biochemistry

What is the normal serum sodium range and what defines hyponatraemia?

Answer

Normal serum sodium is 135 to 145 mmol/L; hyponatraemia is defined as a serum sodium below 135 mmol/L.

#186 · Physiology Anatomy and Biochemistry

What is the risk of correcting chronic hyponatraemia too quickly?

Answer

Rapid correction risks osmotic demyelination syndrome (central pontine myelinolysis); correction should not exceed 8 to 10 mmol/L in 24 hours.

#187 · Physiology Anatomy and Biochemistry

What ECG changes are associated with hyperkalaemia?

Answer

Progressive changes include tented T waves, flattened P waves, widened QRS complexes, and ultimately a sine-wave pattern preceding ventricular fibrillation or asystole.

#188 · Physiology Anatomy and Biochemistry

What is the first-line emergency treatment for severe hyperkalaemia with ECG changes?

Answer

Intravenous calcium gluconate or calcium chloride to stabilise the cardiac membrane, followed by insulin with dextrose to shift potassium intracellularly.

#189 · Physiology Anatomy and Biochemistry

What is the normal serum potassium range?

Answer

Normal serum potassium is 3.5 to 5.0 mmol/L.

#190 · Physiology Anatomy and Biochemistry

What are the clinical features of severe hypokalaemia?

Answer

Muscle weakness, cardiac arrhythmias, ECG changes including U waves and ST depression, and increased sensitivity to non-depolarising neuromuscular blockers.

#191 · Physiology Anatomy and Biochemistry

What is the normal corrected serum calcium range?

Answer

Normal corrected total calcium is 2.20 to 2.60 mmol/L; ionised calcium is normally 1.10 to 1.30 mmol/L.

#192 · Physiology Anatomy and Biochemistry

What ECG change is classically associated with hypercalcaemia?

Answer

A shortened QT interval; severe hypercalcaemia can also cause arrhythmias and, with digoxin, increased toxicity risk.

#193 · Physiology Anatomy and Biochemistry

What are the clinical features of hypocalcaemia?

Answer

Perioral tingling, tetany, Chvostek's and Trousseau's signs, prolonged QT interval, and in severe cases laryngospasm or seizures.

#194 · Physiology Anatomy and Biochemistry

What is the normal serum magnesium range?

Answer

Normal serum magnesium is 0.70 to 1.00 mmol/L.

#195 · Physiology Anatomy and Biochemistry

What are the clinical effects of severe hypermagnesaemia?

Answer

Loss of deep tendon reflexes, hypotension, muscle weakness, respiratory depression, and at very high levels cardiac arrest; it potentiates neuromuscular blockade.

#196 · Physiology Anatomy and Biochemistry

What proportion of cardiac output does the liver receive and from which sources?

Answer

The liver receives about 25% of cardiac output, roughly 75% from the portal vein and 25% from the hepatic artery, though the hepatic artery supplies about half the liver's oxygen delivery.

#197 · Physiology Anatomy and Biochemistry

What is the hepatic arterial buffer response?

Answer

A fall in portal venous flow causes compensatory hepatic arterial vasodilation, mediated by washout of adenosine, to help maintain total hepatic blood flow.

#198 · Physiology Anatomy and Biochemistry

What is meant by first-pass metabolism?

Answer

The extensive hepatic metabolism of an orally absorbed drug before it reaches the systemic circulation, which can markedly reduce bioavailability of drugs such as morphine and propranolol.

#199 · Physiology Anatomy and Biochemistry

Name two synthetic functions of the liver relevant to anaesthesia.

Answer

Synthesis of clotting factors (except factor VIII and von Willebrand factor) and synthesis of albumin and pseudocholinesterase (plasma cholinesterase).

#200 · Physiology Anatomy and Biochemistry

What reactions characterise phase I hepatic drug metabolism?

Answer

Oxidation, reduction and hydrolysis, chiefly via the cytochrome P450 enzyme system, which typically increases drug polarity and may produce active or inactive metabolites.

#201 · Physiology Anatomy and Biochemistry

What reactions characterise phase II hepatic drug metabolism?

Answer

Conjugation reactions such as glucuronidation, sulfation and acetylation, which attach a polar group to increase water solubility and promote renal or biliary excretion.

#202 · Physiology Anatomy and Biochemistry

How does inhalational anaesthesia affect hepatic blood flow?

Answer

Volatile agents generally reduce portal venous flow and total hepatic blood flow in a dose-dependent manner, though halothane reduces it more than modern agents such as sevoflurane and isoflurane.

#203 · Physiology Anatomy and Biochemistry

What are the three phases of the surgical stress response?

Answer

The ebb phase (early hypometabolic phase with reduced cardiac output), the flow phase (hypermetabolic, catabolic phase), and the recovery (anabolic) phase.

#204 · Physiology Anatomy and Biochemistry

Which hormones are characteristically increased during the surgical stress response?

Answer

Cortisol, catecholamines, glucagon, growth hormone and antidiuretic hormone are increased, while insulin secretion and action are relatively reduced.

#205 · Physiology Anatomy and Biochemistry

What metabolic changes occur during the catabolic (flow) phase of the stress response?

Answer

Increased gluconeogenesis and glycogenolysis causing hyperglycaemia, increased proteolysis with negative nitrogen balance, and increased lipolysis.

#206 · Physiology Anatomy and Biochemistry

How can regional or neuraxial anaesthesia modify the surgical stress response?

Answer

Regional and neuraxial techniques blunt the afferent neural pathway of the stress response, reducing cortisol and catecholamine surges, particularly for lower limb and pelvic surgery.

#207 · Physiology Anatomy and Biochemistry

What are the clinical features of thyroid storm?

Answer

Severe hyperthermia, tachycardia or arrhythmias, agitation or altered consciousness, and cardiovascular collapse; it is a life-threatening exacerbation of thyrotoxicosis, often precipitated by surgery or infection.

#208 · Physiology Anatomy and Biochemistry

What is the initial management of thyroid storm?

Answer

Supportive cooling and fluid resuscitation, beta blockade (typically propranolol), antithyroid drugs such as propylthiouracil or carbimazole, and hydrocortisone to reduce peripheral T4 to T3 conversion and cover relative adrenal insufficiency.

#209 · Physiology Anatomy and Biochemistry

What are the clinical features of an Addisonian (adrenal) crisis?

Answer

Profound hypotension refractory to fluids and vasopressors, hypoglycaemia, hyponatraemia, hyperkalaemia, abdominal pain and vomiting.

#210 · Physiology Anatomy and Biochemistry

What is the emergency treatment of an Addisonian crisis?

Answer

Immediate intravenous hydrocortisone, intravenous fluid resuscitation with saline and dextrose for hypoglycaemia, and correction of electrolyte abnormalities.

#211 · Physiology Anatomy and Biochemistry

Why is perioperative steroid cover required in patients on long-term corticosteroids?

Answer

Chronic steroid use suppresses the hypothalamic-pituitary-adrenal axis, impairing the normal cortisol surge needed to meet surgical stress, risking an Addisonian-type crisis without supplementation.

#212 · Physiology Anatomy and Biochemistry

What is a common regimen for perioperative steroid cover in major surgery?

Answer

The patient's usual steroid dose plus intravenous hydrocortisone at induction, with further doses depending on the magnitude of surgery, following current national guidance.

#213 · Physiology Anatomy and Biochemistry

What is the physiological basis of refeeding syndrome?

Answer

Reintroducing carbohydrate after prolonged starvation stimulates insulin release, driving potassium, magnesium and phosphate intracellularly, causing severe extracellular deficiency of these ions.

#214 · Physiology Anatomy and Biochemistry

Which electrolyte abnormality is most characteristic of refeeding syndrome?

Answer

Severe hypophosphataemia, which can cause muscle weakness, respiratory failure, cardiac dysfunction and rhabdomyolysis.

#215 · Physiology Anatomy and Biochemistry

Which patients are at high risk of refeeding syndrome?

Answer

Patients with prolonged starvation, chronic alcoholism, anorexia nervosa, malignancy, or little nutritional intake for more than ten days before refeeding.

#216 · Physiology Anatomy and Biochemistry

How does prolonged starvation change the body's primary metabolic fuel source?

Answer

After initial glycogen depletion, the body shifts to gluconeogenesis and then predominantly to fat metabolism, producing ketone bodies as an alternative fuel for the brain.

#217 · Physiology Anatomy and Biochemistry

How should refeeding be managed to reduce the risk of refeeding syndrome?

Answer

Introduce nutrition slowly starting at a low caloric intake, correct electrolytes before and during feeding, and monitor and replace potassium, magnesium and phosphate closely, alongside thiamine supplementation.

#218 · Physiology Anatomy and Biochemistry

What is the countercurrent multiplier mechanism in the kidney?

Answer

The loop of Henle establishes a progressively hyperosmolar medullary interstitium through active sodium chloride transport, enabling the collecting duct to concentrate urine under antidiuretic hormone influence.

#219 · Physiology Anatomy and Biochemistry

What is the role of erythropoietin in renal physiology?

Answer

Erythropoietin is produced by peritubular fibroblasts in the renal cortex in response to hypoxia and stimulates red blood cell production in the bone marrow.

#220 · Physiology Anatomy and Biochemistry

How is lactate normally cleared and why does impaired clearance matter?

Answer

Lactate is cleared predominantly by hepatic metabolism, with a smaller renal contribution; impaired liver function or shock states reduce clearance and worsen a high anion gap metabolic acidosis.

#221 · Physiology Anatomy and Biochemistry

What is normal cerebral blood flow in the average adult?

Answer

Approximately 50 mL/100g/minute, representing about 15% of resting cardiac output despite the brain being only 2% of body weight.

#222 · Physiology Anatomy and Biochemistry

Over what mean arterial pressure range does cerebral autoregulation normally operate?

Answer

Between approximately 60 and 150 mmHg, over which range cerebral blood flow remains relatively constant through myogenic adjustment of arteriolar tone.

#223 · Physiology Anatomy and Biochemistry

What happens to cerebral autoregulation in chronic hypertension?

Answer

The autoregulatory curve shifts to the right, so higher pressures are needed to maintain flow and previously normal pressures may cause cerebral hypoperfusion.

#224 · Physiology Anatomy and Biochemistry

How does PaCO₂ affect cerebral blood flow?

Answer

Cerebral blood flow changes by roughly 1 to 2 mL/100g/minute per mmHg change in PaCO₂ between about 20 and 80 mmHg, with hypercapnia causing vasodilatation and hypocapnia causing vasoconstriction.

#225 · Physiology Anatomy and Biochemistry

State the Monro-Kellie doctrine.

Answer

The skull is a rigid, non-expansile container holding brain, blood and cerebrospinal fluid in fixed total volume, so an increase in one component must be compensated by a decrease in another or intracranial pressure will rise.

#226 · Physiology Anatomy and Biochemistry

What is the normal range for intracranial pressure in a supine adult?

Answer

Approximately 7 to 15 mmHg; sustained values above 20 mmHg are considered pathological and warrant treatment.

#227 · Physiology Anatomy and Biochemistry

How is cerebral perfusion pressure calculated?

Answer

Cerebral perfusion pressure equals mean arterial pressure minus intracranial pressure (or central venous pressure if higher).

#228 · Physiology Anatomy and Biochemistry

Which two compensatory mechanisms initially buffer a rising intracranial volume?

Answer

Displacement of cerebrospinal fluid into the spinal theca and compression of the low-pressure venous system, both of which are exhausted before intracranial pressure rises steeply.

#229 · Physiology Anatomy and Biochemistry

What defines the resting membrane potential of a typical nerve cell?

Answer

Approximately negative 70 mV, maintained mainly by the outward potassium concentration gradient and the activity of the sodium-potassium ATPase pump.

#230 · Physiology Anatomy and Biochemistry

What triggers the upstroke of an action potential in a nerve fibre?

Answer

Depolarisation past threshold opens voltage-gated sodium channels, causing a rapid influx of sodium ions and reversal of membrane polarity.

#231 · Physiology Anatomy and Biochemistry

Why does conduction velocity increase with axon diameter?

Answer

A larger diameter reduces internal axial resistance to current flow, allowing local circuit currents to depolarise adjacent membrane more rapidly.

#232 · Physiology Anatomy and Biochemistry

What is saltatory conduction?

Answer

In myelinated fibres, the action potential jumps from one node of Ranvier to the next because myelin insulates the internode, greatly increasing conduction speed.

#233 · Physiology Anatomy and Biochemistry

According to Erlanger and Gasser classification, which fibre type is most sensitive to local anaesthetic block?

Answer

Small myelinated B fibres (preganglionic autonomic) and small C fibres are generally blocked before larger A fibres, though clinical practice shows overlap rather than a strict sequence.

#234 · Physiology Anatomy and Biochemistry

What is the neurotransmitter released at the neuromuscular junction?

Answer

Acetylcholine, released from vesicles in the motor nerve terminal following calcium influx triggered by nerve terminal depolarisation.

#235 · Physiology Anatomy and Biochemistry

Which receptor does acetylcholine act on at the postjunctional membrane of skeletal muscle?

Answer

The nicotinic acetylcholine receptor, a ligand-gated ion channel whose opening allows sodium influx and potassium efflux, generating an end-plate potential.

#236 · Physiology Anatomy and Biochemistry

How do non-depolarising neuromuscular blockers work?

Answer

They competitively antagonise acetylcholine at the nicotinic receptor without activating it, preventing generation of an end-plate potential and producing flaccid paralysis.

#237 · Physiology Anatomy and Biochemistry

Why does suxamethonium cause fasciculations before paralysis?

Answer

As a depolarising agent it initially opens the nicotinic receptor channel like acetylcholine, causing widespread muscle fibre depolarisation and fasciculation before persistent depolarisation blocks further transmission.

#238 · Physiology Anatomy and Biochemistry

Which dermatome corresponds to the thumb?

Answer

C6, a key landmark for assessing the level of an upper limb block or high neuraxial block.

#239 · Physiology Anatomy and Biochemistry

Which dermatome corresponds to the nipple line?

Answer

T4, commonly used as the target sensory level for regional anaesthesia for caesarean section.

#240 · Physiology Anatomy and Biochemistry

Which dermatome corresponds to the umbilicus?

Answer

T10, a useful landmark confirming adequate block height for procedures such as lower abdominal surgery or labour analgesia.

#241 · Physiology Anatomy and Biochemistry

Which dermatome corresponds to the inguinal or groin region?

Answer

L1, marking the lower thoracic to upper lumbar transition tested when assessing spinal or epidural block spread.

#242 · Physiology Anatomy and Biochemistry

Which dermatome corresponds to the lateral border of the foot and little toe?

Answer

S1, useful for confirming sacral spread of a spinal or epidural block, which can be the last region to become blocked.

#243 · Physiology Anatomy and Biochemistry

Which dermatome supplies the little finger?

Answer

C8, adjacent to but distinct from the C6 thumb and C7 middle finger landmarks used to map the upper limb dermatomal sequence.

#244 · Physiology Anatomy and Biochemistry

Which spinal roots contribute to the brachial plexus?

Answer

The anterior rami of C5, C6, C7, C8 and T1, sometimes with small contributions from C4 or T2.

#245 · Physiology Anatomy and Biochemistry

State the order of structures in the brachial plexus from proximal to distal.

Answer

Roots, trunks, divisions, cords, then terminal branches, commonly remembered by the mnemonic Real Teenagers Drink Cold Beer.

#246 · Physiology Anatomy and Biochemistry

How many trunks form the brachial plexus and from which roots?

Answer

Three trunks: the upper trunk from C5 and C6, the middle trunk from C7 alone, and the lower trunk from C8 and T1.

#247 · Physiology Anatomy and Biochemistry

How are the three cords of the brachial plexus named and formed?

Answer

The posterior cord forms from the posterior divisions of all three trunks, the lateral cord from the anterior divisions of the upper and middle trunks, and the medial cord from the anterior division of the lower trunk alone.

#248 · Physiology Anatomy and Biochemistry

Which cords give rise to the median nerve?

Answer

The median nerve arises from a lateral root off the lateral cord and a medial root off the medial cord, which join anterior to the axillary artery.

#249 · Physiology Anatomy and Biochemistry

From which cord does the radial nerve arise?

Answer

The posterior cord, which continues as the radial nerve after giving off the axillary nerve.

#250 · Physiology Anatomy and Biochemistry

From which cord does the ulnar nerve arise?

Answer

The medial cord, as one of its two terminal branches alongside the medial contribution to the median nerve.

#251 · Physiology Anatomy and Biochemistry

Which nerve is the sole terminal branch of the lateral cord?

Answer

The musculocutaneous nerve, which supplies the flexor muscles of the anterior arm and continues as the lateral cutaneous nerve of the forearm.

#252 · Physiology Anatomy and Biochemistry

Which nerve arises from the posterior cord alongside the radial nerve?

Answer

The axillary nerve, which supplies deltoid and teres minor and gives cutaneous sensation over the regimental badge area of the shoulder.

#253 · Physiology Anatomy and Biochemistry

At which level of the brachial plexus does an interscalene block act?

Answer

At the level of the roots or upper trunks between the scalenus anterior and scalenus medius muscles, providing good shoulder and proximal arm anaesthesia but often sparing the C8 to T1 territory.

#254 · Physiology Anatomy and Biochemistry

At which level of the brachial plexus does a supraclavicular block act?

Answer

At the level of the trunks or divisions as they pass over the first rib close to the subclavian artery, giving dense blockade of the whole arm.

#255 · Physiology Anatomy and Biochemistry

At which level of the brachial plexus does an infraclavicular block act?

Answer

At the level of the cords, deep to pectoralis minor, providing reliable coverage for forearm and hand surgery.

#256 · Physiology Anatomy and Biochemistry

At which level of the brachial plexus does an axillary block act?

Answer

At the level of the terminal branches around the axillary artery, blocking the median, ulnar, radial and musculocutaneous nerves individually and typically sparing the intercostobrachial nerve.

#257 · Physiology Anatomy and Biochemistry

Which laryngeal cartilage is the largest and forms the laryngeal prominence?

Answer

The thyroid cartilage, a shield-shaped structure whose anterior fusion forms the laryngeal prominence, more pronounced in males.

#258 · Physiology Anatomy and Biochemistry

Which laryngeal cartilage is the only complete ring in the airway?

Answer

The cricoid cartilage, which forms a complete signet-ring shape and lies below the thyroid cartilage at the level of C6.

#259 · Physiology Anatomy and Biochemistry

What is the function of the paired arytenoid cartilages?

Answer

They sit on the posterior cricoid lamina and rotate and glide to alter vocal cord tension and position, controlling phonation and glottic opening.

#260 · Physiology Anatomy and Biochemistry

What is the epiglottis and where does it attach?

Answer

A leaf-shaped elastic cartilage attached to the inner surface of the thyroid cartilage that folds over the laryngeal inlet during swallowing to protect the airway.

#261 · Physiology Anatomy and Biochemistry

Where is the cricothyroid membrane located and why is it clinically important?

Answer

It spans between the lower border of the thyroid cartilage and the upper border of the cricoid cartilage in the midline, and is the standard site for emergency front-of-neck airway access.

#262 · Physiology Anatomy and Biochemistry

Which nerve provides sensory innervation to the airway above the vocal cords?

Answer

The internal branch of the superior laryngeal nerve, a branch of the vagus, supplies mucosal sensation from the epiglottis down to the level of the vocal cords.

#263 · Physiology Anatomy and Biochemistry

Which nerve provides sensory innervation to the airway below the vocal cords?

Answer

The recurrent laryngeal nerve, another branch of the vagus, supplies sensation to the subglottic mucosa and trachea and also motor supply to all intrinsic laryngeal muscles except cricothyroid.

#264 · Physiology Anatomy and Biochemistry

Which nerve is anaesthetised to block sensation over the base of tongue and vallecula for awake fiberoptic intubation?

Answer

The glossopharyngeal nerve, typically blocked at the palatoglossal fold or by topicalisation, providing sensation to the posterior third of the tongue and oropharynx.

#265 · Physiology Anatomy and Biochemistry

How is a superior laryngeal nerve block typically performed for awake intubation?

Answer

Local anaesthetic is injected just below the greater cornu of the hyoid bone, where the nerve pierces the thyrohyoid membrane, anaesthetising the supraglottic mucosa.

#266 · Physiology Anatomy and Biochemistry

What structure lies immediately superficial to the dura mater in the epidural space?

Answer

The ligamentum flavum, a dense elastic ligament connecting adjacent vertebral laminae, whose characteristic increased resistance is felt during loss-of-resistance technique before entering the epidural space.

#267 · Physiology Anatomy and Biochemistry

What is the composition of the ligamentum flavum and why does it matter clinically?

Answer

It is composed largely of elastic fibres rather than collagen, giving it a distinct dense, springy feel on needle advancement that helps identify the loss-of-resistance point of the epidural space.

#268 · Physiology Anatomy and Biochemistry

At what vertebral level does the adult spinal cord typically terminate as the conus medullaris?

Answer

Approximately the L1 to L2 vertebral level in adults, though it may end as high as T12 or as low as L3.

#269 · Physiology Anatomy and Biochemistry

Why is a lumbar puncture or spinal anaesthetic usually performed at L3 to L4 or L4 to L5?

Answer

Below the conus medullaris the space contains only the cauda equina, so a needle inserted at these lower interspaces avoids direct spinal cord injury.

#270 · Physiology Anatomy and Biochemistry

What landmark is commonly used to identify the L4 spinous process or L3 to L4 interspace?

Answer

Tuffier's line, an imaginary line joining the highest points of the iliac crests, which typically crosses the L4 spinous process or the L3 to L4 interspace.

#271 · Physiology Anatomy and Biochemistry

Where does the dural sac usually terminate in adults?

Answer

At approximately the S2 vertebral level, marking the lower limit of the subarachnoid space and a boundary relevant to caudal epidural technique.

#272 · Physiology Anatomy and Biochemistry

Which anatomical landmarks guide ultrasound-guided internal jugular vein cannulation?

Answer

The vein lies lateral to the carotid artery within the carotid sheath, typically deep to the triangle formed by the sternal and clavicular heads of sternocleidomastoid and the clavicle.

#273 · Physiology Anatomy and Biochemistry

What is the main anatomical risk associated with subclavian vein cannulation?

Answer

The subclavian artery and the apex of the lung lie close behind the vein, so an inaccurate approach carries risk of arterial puncture or pneumothorax.

#274 · Physiology Anatomy and Biochemistry

Where does the femoral vein lie in relation to the femoral artery at the inguinal ligament?

Answer

Medial to the femoral artery, itself medial to the femoral nerve, remembered by the mnemonic NAVEL from lateral to medial: Nerve, Artery, Vein, Empty space, Lymphatics.

#275 · Physiology Anatomy and Biochemistry

Why is the right internal jugular vein generally preferred over the left for central venous catheterisation?

Answer

It provides a straighter, more direct path to the superior vena cava and right atrium, and avoids proximity to the thoracic duct which lies on the left side of the neck.

#276 · Physics Clinical Measurement and Data Interpretation

State Boyle's Law.

Answer

At constant temperature, the volume of a fixed mass of gas is inversely proportional to its pressure. This is expressed as $PV = k$, or $P_1V_1 = P_2V_2$.

#277 · Physics Clinical Measurement and Data Interpretation

State Charles's Law.

Answer

At constant pressure, the volume of a fixed mass of gas is directly proportional to its absolute temperature. This is expressed as $V/T = k$.

#278 · Physics Clinical Measurement and Data Interpretation

State Gay-Lussac's Law (the third gas law).

Answer

At constant volume, the pressure of a fixed mass of gas is directly proportional to its absolute temperature. This is expressed as $P/T = k$.

#279 · Physics Clinical Measurement and Data Interpretation

Give the universal (ideal) gas law equation.

Answer

$PV = nRT$, where P is pressure, V is volume, n is the number of moles, R is the universal gas constant and T is absolute temperature in kelvin.

#280 · Physics Clinical Measurement and Data Interpretation

What is the value of the universal gas constant R?

Answer

R equals 8.314 joules per mole per kelvin (J/mol/K).

#281 · Physics Clinical Measurement and Data Interpretation

Define critical temperature of a gas.

Answer

The critical temperature is the temperature above which a gas cannot be liquefied, no matter how much pressure is applied.

#282 · Physics Clinical Measurement and Data Interpretation

Define critical pressure of a gas.

Answer

The critical pressure is the pressure required to liquefy a gas when it is exactly at its critical temperature.

#283 · Physics Clinical Measurement and Data Interpretation

What is the critical temperature of oxygen?

Answer

The critical temperature of oxygen is minus 118 degrees Celsius, so oxygen cannot exist as a liquid at normal ambient temperature and is stored as a compressed gas.

#284 · Physics Clinical Measurement and Data Interpretation

What is the critical temperature of nitrous oxide?

Answer

The critical temperature of nitrous oxide is 36.5 degrees Celsius, which is close to normal room temperature, so it is stored as a liquid with vapour above it in a cylinder.

#285 · Physics Clinical Measurement and Data Interpretation

Why does a nitrous oxide cylinder pressure gauge stay constant as the cylinder empties, until nearly empty?

Answer

While liquid nitrous oxide remains in the cylinder, the pressure reflects the saturated vapour pressure above the liquid and stays constant. Only once all the liquid has evaporated does the pressure begin to fall as the remaining gas is used.

#286 · Physics Clinical Measurement and Data Interpretation

How can you estimate the amount of nitrous oxide remaining in a cylinder?

Answer

Because pressure does not fall until the liquid is exhausted, the contents must be estimated by weighing the cylinder and subtracting the known tare (empty) weight.

#287 · Physics Clinical Measurement and Data Interpretation

What is Entonox?

Answer

Entonox is a premixed cylinder gas containing 50% nitrous oxide and 50% oxygen by volume, used for analgesia.

#288 · Physics Clinical Measurement and Data Interpretation

What is the pseudocritical temperature of Entonox and why does it matter?

Answer

The pseudocritical temperature of Entonox is minus 7 degrees Celsius. Below this temperature the gases can separate (lamination), so oxygen-poor gas may be delivered from the bottom of the cylinder, risking a hypoxic mixture.

#289 · Physics Clinical Measurement and Data Interpretation

How should a cold Entonox cylinder be managed before use?

Answer

It should be warmed to above minus 7 degrees Celsius and inverted or agitated several times to remix any separated liquid nitrous oxide with the oxygen before use.

#290 · Physics Clinical Measurement and Data Interpretation

What is the colour coding of a UK oxygen cylinder?

Answer

The oxygen cylinder is white throughout (body and shoulder) under the current UK/EU standard; this replaced the older black body/white shoulder scheme.

#291 · Physics Clinical Measurement and Data Interpretation

What is the colour coding of a UK nitrous oxide cylinder?

Answer

The nitrous oxide cylinder has a white body with a French blue shoulder under the current standard.

#292 · Physics Clinical Measurement and Data Interpretation

What is the colour coding of an Entonox cylinder?

Answer

The Entonox cylinder has a white body with a blue and white quartered shoulder.

#293 · Physics Clinical Measurement and Data Interpretation

What is the SI unit of pressure?

Answer

The pascal (Pa), defined as one newton per square metre. In clinical practice, kilopascals (kPa) are the standard unit for arterial blood gas values.

#294 · Physics Clinical Measurement and Data Interpretation

Convert 1 kilopascal (kPa) to mmHg.

Answer

1 kPa is approximately equal to 7.5 mmHg.

#295 · Physics Clinical Measurement and Data Interpretation

Convert 1 atmosphere to kPa.

Answer

1 atmosphere is approximately 101.3 kPa.

#296 · Physics Clinical Measurement and Data Interpretation

Convert 1 atmosphere to mmHg.

Answer

1 atmosphere is approximately 760 mmHg.

#297 · Physics Clinical Measurement and Data Interpretation

Convert 1 bar to kPa.

Answer

1 bar is equal to 100 kPa, and approximately equal to 1 atmosphere.

#298 · Physics Clinical Measurement and Data Interpretation

Convert 1 kPa to cmH₂O.

Answer

1 kPa is approximately equal to 10.2 cmH₂O.

#299 · Physics Clinical Measurement and Data Interpretation

Why is central venous pressure often measured in cmH₂O rather than mmHg?

Answer

cmH₂O gives finer resolution for the low pressures found in the venous system, since 1 mmHg is roughly 1.36 cmH₂O, giving more usable numbers at that scale.

#300 · Physics Clinical Measurement and Data Interpretation

What is the typical filling pressure of a full UK oxygen cylinder?

Answer

A full oxygen cylinder is typically pressurised to around 137 bar (approximately 13,700 kPa) at 15 degrees Celsius.

#301 · Physics Clinical Measurement and Data Interpretation

Distinguish laminar flow from turbulent flow.

Answer

Laminar flow is smooth, with fluid moving in parallel layers at a velocity profile that is fastest in the centre. Turbulent flow is chaotic and disorganised, with eddies and cross-currents, and occurs at higher flow rates or velocities.

#302 · Physics Clinical Measurement and Data Interpretation

State the Hagen-Poiseuille equation for laminar flow through a tube.

Answer

Flow equals (π multiplied by pressure difference multiplied by radius to the fourth power) divided by (8 multiplied by viscosity multiplied by length). Flow is therefore proportional to the fourth power of the radius.

#303 · Physics Clinical Measurement and Data Interpretation

What happens to laminar flow through a tube if the radius is halved, all else being equal?

Answer

Flow falls to one-sixteenth of its original value, since flow is proportional to the radius raised to the fourth power.

#304 · Physics Clinical Measurement and Data Interpretation

In laminar flow, what determines the driving pressure needed for a given flow?

Answer

Driving pressure is proportional to flow and to viscosity, and inversely proportional to the fourth power of the radius, as described by the Hagen-Poiseuille equation.

#305 · Physics Clinical Measurement and Data Interpretation

In turbulent flow, what determines the driving pressure needed for a given flow?

Answer

In turbulent flow, driving pressure is proportional to the square of the flow rate and to the density of the fluid, rather than to viscosity as in laminar flow.

#306 · Physics Clinical Measurement and Data Interpretation

Define the Reynolds number.

Answer

The Reynolds number is a dimensionless value that predicts whether flow will be laminar or turbulent, calculated from velocity, density, tube diameter and viscosity of the fluid.

#307 · Physics Clinical Measurement and Data Interpretation

What Reynolds number threshold suggests a transition from laminar to turbulent flow?

Answer

A Reynolds number below approximately 2000 predicts laminar flow, and above approximately 4000 predicts turbulent flow; values in between are an unpredictable transitional zone.

#308 · Physics Clinical Measurement and Data Interpretation

Which factors increase the likelihood of turbulent rather than laminar flow?

Answer

High flow velocity, large tube diameter, low fluid viscosity, high fluid density, and any sharp change in tube diameter or direction all favour turbulent flow.

#309 · Physics Clinical Measurement and Data Interpretation

Why does gas flow through a narrow tracheal tube or airway obstruction tend to become turbulent?

Answer

A sudden reduction in diameter increases flow velocity through the narrowed segment, which raises the Reynolds number and promotes turbulent flow.

#310 · Physics Clinical Measurement and Data Interpretation

Why is heliox sometimes used in severe upper airway obstruction?

Answer

Helium has a much lower density than nitrogen, and since turbulent flow depends on density, a helium-oxygen mixture reduces turbulence and can improve gas flow past an obstruction.

#311 · Physics Clinical Measurement and Data Interpretation

How does a variable-orifice rotameter flowmeter work?

Answer

Gas flows upward through a tapered glass tube, lifting a bobbin or float until the annular gap around it is wide enough that gas escaping around it balances the gravitational pull on the float, with the float position indicating flow.

#312 · Physics Clinical Measurement and Data Interpretation

Why does flow past the bobbin change character as it rises up a rotameter tube?

Answer

At the bottom of the tube the annular gap is narrow and flow is laminar, dependent on gas viscosity. Higher up, the wider gap produces turbulent flow, dependent on gas density.

#313 · Physics Clinical Measurement and Data Interpretation

Why must each rotameter tube and bobbin be calibrated for a specific gas?

Answer

Because flow through the tube depends on the viscosity and density of the particular gas, a tube calibrated for oxygen would give an inaccurate reading if used with a different gas such as nitrous oxide.

#314 · Physics Clinical Measurement and Data Interpretation

Why are rotameter bobbins read at different points depending on their shape?

Answer

A ball-shaped float is read at its widest, horizontal diameter (the equator), whereas a bobbin with a skirt is read at the top edge of the skirt.

#315 · Physics Clinical Measurement and Data Interpretation

Why are rotameter tubes mounted vertically and slightly conical?

Answer

The tube must be vertical so gravity acts consistently on the float, and the taper widens the annular gap progressively with height, allowing the tube to display a wide range of flows on one scale.

#316 · Physics Clinical Measurement and Data Interpretation

What is the splitting ratio in a plenum vaporiser?

Answer

The splitting ratio is the ratio of fresh gas flow diverted through the vaporising chamber (where it becomes saturated with vapour) to the fresh gas flow that bypasses the chamber, which together determine the final delivered concentration.

#317 · Physics Clinical Measurement and Data Interpretation

What property of a volatile agent determines the splitting ratio needed for a given vaporiser output?

Answer

The saturated vapour pressure of the agent at a given temperature determines the splitting ratio, since an agent with a higher vapour pressure needs a smaller fraction of gas diverted through the vaporising chamber to achieve the same output concentration.

#318 · Physics Clinical Measurement and Data Interpretation

Why do plenum vaporisers need temperature compensation?

Answer

Vaporisation is an endothermic process, so the vaporising chamber cools as liquid agent evaporates, which would reduce its vapour pressure and output concentration unless compensated for.

#319 · Physics Clinical Measurement and Data Interpretation

How does a bimetallic strip provide temperature compensation in a vaporiser?

Answer

A bimetallic strip, made of two metals with different thermal expansion coefficients, bends as temperature falls and narrows the bypass channel, diverting a greater proportion of fresh gas through the vaporising chamber to compensate for the falling saturated vapour pressure and keep output concentration stable.

#320 · Physics Clinical Measurement and Data Interpretation

Why must a plenum vaporiser be agent-specific?

Answer

Because each volatile agent has a different saturated vapour pressure and required splitting ratio at any given temperature, a vaporiser calibrated for one agent will deliver an inaccurate concentration if filled with a different agent.

#321 · Physics Clinical Measurement and Data Interpretation

Define macroshock.

Answer

Macroshock is an electric shock caused by current passing through intact skin, typically requiring more than about 1 milliamp to be perceived and larger currents to cause harm such as ventricular fibrillation.

#322 · Physics Clinical Measurement and Data Interpretation

Define microshock.

Answer

Microshock is a small current, as little as 50 to 100 microamps, applied directly to or near the heart (for example via a central venous catheter or pacing wire), which bypasses the skin's resistance and can cause ventricular fibrillation.

#323 · Physics Clinical Measurement and Data Interpretation

Why is the threshold for microshock so much lower than for macroshock?

Answer

Skin provides significant electrical resistance that normally limits current reaching the heart, but a direct intracardiac pathway bypasses this protection, so a much smaller current delivered directly to the myocardium can trigger fibrillation.

#324 · Physics Clinical Measurement and Data Interpretation

What is the purpose of earthing (grounding) electrical equipment?

Answer

Earthing provides a low-resistance path to the ground for any fault current, so that in the event of an insulation failure, current flows safely to earth rather than through a person touching the equipment.

#325 · Physics Clinical Measurement and Data Interpretation

What is an isolated (floating) power supply and why is it used in operating theatres?

Answer

An isolated power supply has no direct connection to earth, so that if a single fault causes one live conductor to contact the earthed casing, no large fault current flows and the circuit can keep functioning safely until the fault is found.

#326 · Physics Clinical Measurement and Data Interpretation

Describe Type B equipment under the IEC medical electrical equipment classification.

Answer

Type B equipment provides protection against electric shock with an earthed enclosure and permissible leakage current, but its applied parts are not suitable for direct connection to the heart.

#327 · Physics Clinical Measurement and Data Interpretation

Describe Type BF equipment.

Answer

Type BF equipment has an isolated (floating) applied part, giving a higher degree of protection than Type B, but it is still not intended for direct cardiac connection.

#328 · Physics Clinical Measurement and Data Interpretation

Describe Type CF equipment.

Answer

Type CF equipment provides the highest level of protection, with an isolated applied part and very low permissible leakage current, making it suitable for direct connection to the heart, such as ECG leads or invasive pressure lines.

#329 · Physics Clinical Measurement and Data Interpretation

Distinguish monopolar from bipolar diathermy.

Answer

In monopolar diathermy, current passes from a small active electrode through the patient's body to a large return (neutral) electrode elsewhere on the skin. In bipolar diathermy, current passes only between the two tips of forceps, through a small volume of tissue, with no separate return electrode needed.

#330 · Physics Clinical Measurement and Data Interpretation

Why must the return electrode in monopolar diathermy have a large surface area and good contact?

Answer

A large, well-applied contact area spreads the current density over a wide skin surface so it does not generate enough heat to cause a burn, whereas poor contact concentrates current and risks a diathermy burn at that site.

#331 · Physics Clinical Measurement and Data Interpretation

What colour is an oxygen cylinder in the UK?

Answer

White shoulders and body under the current ISO 32 standard, which replaced the older black body/white shoulder colour scheme.

#332 · Physics Clinical Measurement and Data Interpretation

What colour is a nitrous oxide cylinder?

Answer

White body with a French blue shoulder under the current standard (the body changed from blue to white; only the shoulder colour is retained).

#333 · Physics Clinical Measurement and Data Interpretation

What colour is an air cylinder for medical use?

Answer

Black and white quartered shoulders with a white body under the current ISO 32 standard.

#334 · Physics Clinical Measurement and Data Interpretation

What colour is an Entonox cylinder?

Answer

Blue and white quartered shoulders with a white body, since Entonox is a 50:50 mix of oxygen and nitrous oxide.

#335 · Physics Clinical Measurement and Data Interpretation

What colour is a carbon dioxide cylinder?

Answer

Grey shoulders with a white body, used mainly for laparoscopic insufflation rather than anaesthetic delivery.

#336 · Physics Clinical Measurement and Data Interpretation

What is the purpose of the pin-index safety system?

Answer

It uses a unique arrangement of two pins on the cylinder yoke matching holes on the cylinder valve for each gas, preventing the wrong cylinder being fitted to a yoke.

#337 · Physics Clinical Measurement and Data Interpretation

What is the Schrader valve system used for?

Answer

It is a non-interchangeable, colour-coded socket and probe system connecting pipeline gas hoses to wall outlets, preventing accidental cross-connection of gases.

#338 · Physics Clinical Measurement and Data Interpretation

What pipeline pressure does UK hospital medical gas supply typically deliver?

Answer

Approximately 4 bar (400 kPa), which is regulated down further within the anaesthetic machine before reaching the flowmeters.

#339 · Physics Clinical Measurement and Data Interpretation

What is the function of a pressure regulator on an anaesthetic machine?

Answer

It reduces the high and variable pressure of a cylinder to a constant, safe working pressure suitable for the flowmeters and downstream circuitry.

#340 · Physics Clinical Measurement and Data Interpretation

Why do modern anaesthetic machine flowmeters use electronic flow sensors rather than rotameters in many designs?

Answer

Electronic flow sensors give more accurate low-flow measurement, are unaffected by static electricity or tube position, and interface directly with the machine's gas mixing and alarm systems.

#341 · Physics Clinical Measurement and Data Interpretation

What is a rotameter and how does it measure gas flow?

Answer

A rotameter is a tapered glass tube containing a bobbin that rises as gas flow increases, the annular gap around the bobbin widening so flow reading is proportional to bobbin position.

#342 · Physics Clinical Measurement and Data Interpretation

Why must oxygen always be added downstream of other gases in a rotameter bank?

Answer

Placing oxygen last, nearest the common gas outlet, minimises the risk of a hypoxic mixture being delivered if a leak occurs upstream in the bank.

#343 · Physics Clinical Measurement and Data Interpretation

What is the common gas outlet?

Answer

The single point on the anaesthetic machine where the mixed fresh gas flow leaves the machine to enter the breathing system.

#344 · Physics Clinical Measurement and Data Interpretation

What safety feature prevents delivery of a hypoxic gas mixture from modern anaesthetic machines?

Answer

An anti-hypoxic device mechanically links nitrous oxide and oxygen flow controls so the fresh gas oxygen concentration cannot fall below approximately 21 to 25 percent.

#345 · Physics Clinical Measurement and Data Interpretation

What triggers the oxygen failure warning device on an anaesthetic machine?

Answer

A fall in oxygen supply pressure below a threshold (around 200 kPa) activates an audible alarm and, in most machines, automatically cuts off nitrous oxide flow.

#346 · Physics Clinical Measurement and Data Interpretation

What is the Bourdon gauge used for on an anaesthetic machine?

Answer

It measures cylinder or pipeline gas pressure using a coiled tube that straightens slightly as pressure rises, moving a needle across a calibrated dial.

#347 · Physics Clinical Measurement and Data Interpretation

Classify the Mapleson A breathing system's efficiency for spontaneous versus controlled ventilation.

Answer

Mapleson A is the most efficient system for spontaneous ventilation but is inefficient for controlled ventilation, requiring high fresh gas flows to prevent rebreathing.

#348 · Physics Clinical Measurement and Data Interpretation

Which Mapleson system is the Bain circuit and what is its configuration?

Answer

The Bain circuit is a coaxial version of the Mapleson D system, with fresh gas delivered through an inner tube running inside the corrugated expiratory limb.

#349 · Physics Clinical Measurement and Data Interpretation

Which Mapleson system is most efficient for controlled ventilation and why?

Answer

Mapleson D (and the Bain modification) is most efficient for controlled ventilation because fresh gas flow flushes dead space gas away from the patient during expiration.

#350 · Physics Clinical Measurement and Data Interpretation

What are the key components of the Mapleson E system?

Answer

The Ayre's T-piece has no reservoir bag or valves, consisting simply of fresh gas delivery, a T-connector at the patient, and an expiratory limb, historically used in paediatric anaesthesia.

#351 · Physics Clinical Measurement and Data Interpretation

What distinguishes the Mapleson F system from the Mapleson E?

Answer

Mapleson F (the Jackson-Rees modification) adds an open-ended reservoir bag to the end of the expiratory limb of the T-piece, allowing manual ventilation.

#352 · Physics Clinical Measurement and Data Interpretation

In the Mapleson A system, where is the adjustable pressure-limiting valve positioned?

Answer

The APL valve sits at the patient end of the system, near the facemask connection, which is why it is efficient for spontaneous ventilation.

#353 · Physics Clinical Measurement and Data Interpretation

What fresh gas flow is generally required to prevent rebreathing with a Mapleson D system during spontaneous ventilation?

Answer

Approximately 2 to 3 times minute volume is typically required, making Mapleson D relatively inefficient for spontaneous ventilation.

#354 · Physics Clinical Measurement and Data Interpretation

What is the main advantage of a circle breathing system over Mapleson systems?

Answer

The circle system allows very low fresh gas flows through carbon dioxide absorption and gas recirculation, reducing volatile agent and fresh gas costs and preserving heat and humidity.

#355 · Physics Clinical Measurement and Data Interpretation

What are the essential components of a circle breathing system?

Answer

Inspiratory and expiratory unidirectional valves, a carbon dioxide absorber, a reservoir bag, an adjustable pressure-limiting valve, and fresh gas inflow, all arranged in a circular pathway.

#356 · Physics Clinical Measurement and Data Interpretation

What is the main active ingredient in soda lime and what does it react with?

Answer

Soda lime is mainly calcium hydroxide with a smaller amount of sodium hydroxide as a catalyst, and it reacts chemically with carbon dioxide to remove it from exhaled gas.

#357 · Physics Clinical Measurement and Data Interpretation

Write the overall chemical reaction of carbon dioxide absorption by soda lime.

Answer

Carbon dioxide plus water forms carbonic acid, which reacts with the hydroxides to form calcium carbonate and sodium carbonate, releasing heat and water as by-products.

#358 · Physics Clinical Measurement and Data Interpretation

Why is soda lime absorption described as an exothermic reaction, and why does this matter clinically?

Answer

It releases heat as carbon dioxide is neutralised, which helps warm inspired gas but can also mean a hot canister signals active, effective absorption is occurring.

#359 · Physics Clinical Measurement and Data Interpretation

What pH indicator dye is commonly added to soda lime and what colour change indicates exhaustion?

Answer

Ethyl violet is commonly used, changing from white to violet or purple as the soda lime becomes exhausted and more acidic, signalling the granules need replacing.

#360 · Physics Clinical Measurement and Data Interpretation

Why has compound A production with sevoflurane and desiccated soda lime become less clinically emphasised with modern absorbents?

Answer

Newer absorbents containing little or no strong alkali (such as calcium hydroxide lime without sodium or potassium hydroxide) produce far less compound A and avoid degradation reactions seen with dried older absorbents.

#361 · Physics Clinical Measurement and Data Interpretation

What is the typical granule size range used for soda lime and why does this matter?

Answer

Granules are usually 4 to 8 mesh size, balancing a large surface area for efficient carbon dioxide absorption against an acceptable resistance to gas flow.

#362 · Physics Clinical Measurement and Data Interpretation

What is the correct cuff pressure range for a tracheal tube cuff?

Answer

Cuff pressure should be maintained between 20 and 30 cmH₂O, high enough to seal the airway but low enough to avoid tracheal mucosal ischaemia.

#363 · Physics Clinical Measurement and Data Interpretation

What complication results from persistently excessive tracheal tube cuff pressure?

Answer

Mucosal capillary perfusion pressure is exceeded, causing ischaemia that can progress to tracheal ulceration, stenosis, or tracheo-oesophageal fistula.

#364 · Physics Clinical Measurement and Data Interpretation

What is the main advantage of videolaryngoscopy over direct laryngoscopy?

Answer

Videolaryngoscopy provides an indirect camera view of the glottis, improving glottic visualisation without requiring alignment of oral, pharyngeal and laryngeal axes, which is particularly useful in difficult airways.

#365 · Physics Clinical Measurement and Data Interpretation

What is the Macintosh laryngoscope blade designed to do?

Answer

It is a curved blade whose tip sits in the vallecula, indirectly lifting the epiglottis forward via the hyoepiglottic ligament to expose the glottis.

#366 · Physics Clinical Measurement and Data Interpretation

How does a supraglottic airway device differ from a tracheal tube in its position?

Answer

A supraglottic airway sits above the glottis, forming a seal around or above the laryngeal inlet, rather than passing through the vocal cords into the trachea.

#367 · Physics Clinical Measurement and Data Interpretation

What is a key advantage of a second-generation supraglottic airway device such as the LMA ProSeal?

Answer

It incorporates a separate drain tube allowing gastric tube passage and providing some protection against aspiration of regurgitated gastric contents.

#368 · Physics Clinical Measurement and Data Interpretation

What is the Cormack and Lehane grading system used for?

Answer

It grades the laryngoscopic view obtained at direct laryngoscopy from Grade 1 (full glottis visible) to Grade 4 (no laryngeal structures visible), predicting intubation difficulty.

#369 · Physics Clinical Measurement and Data Interpretation

What monitoring must be applied before induction of anaesthesia according to AAGBI minimum monitoring standards?

Answer

Pulse oximetry, non-invasive blood pressure, and electrocardiography must all be established and functioning before induction of any general or regional anaesthetic.

#370 · Physics Clinical Measurement and Data Interpretation

What monitor is mandatory to confirm correct tracheal tube placement, per AAGBI standards?

Answer

Capnography (end-tidal carbon dioxide monitoring) is mandatory to confirm tracheal intubation and must be used continuously throughout every general anaesthetic.

#371 • Physics Clinical Measurement and Data Interpretation

What additional monitors does the AAGBI recommend whenever a breathing system or ventilator is used?

Answer

An oxygen analyser with a low-concentration alarm, an airway pressure monitor with disconnection/high-pressure alarms, and a vapour analyser when volatile agents are used.

#372 • Physics Clinical Measurement and Data Interpretation

Who is required to be present throughout every general anaesthetic under AAGBI guidance?

Answer

A trained anaesthetist must be continuously present throughout the conduct of every general anaesthetic and until the patient has recovered satisfactorily.

#373 · Physics Clinical Measurement and Data Interpretation

What additional monitoring should be considered for patients under sedation or regional anaesthesia per AAGBI standards?

Answer

The same minimum standards (pulse oximetry, non-invasive blood pressure, ECG as indicated) apply, with capnography strongly recommended whenever sedation carries a risk of airway compromise.

#374 · Physics Clinical Measurement and Data Interpretation

State the Beer-Lambert law as applied to pulse oximetry.

Answer

It relates the absorption of light passing through a substance to the concentration of the absorbing substance and the path length, allowing calculation of hemoglobin oxygen saturation from light absorption at two wavelengths.

#375 · Physics Clinical Measurement and Data Interpretation

Which two wavelengths of light are typically used in pulse oximetry, and why?

Answer

Red light (approximately 660 nm) and infrared light (approximately 940 nm) are used because oxyhaemoglobin and deoxyhaemoglobin absorb these wavelengths differently, allowing their ratio to be calculated.

#376 · Physics Clinical Measurement and Data Interpretation

How does pulse oximetry distinguish arterial blood signal from venous and tissue absorption?

Answer

It isolates the pulsatile (AC) component of light absorption, which arises from arterial pulsation, from the constant (DC) component caused by venous blood, tissue and bone.

#377 · Physics Clinical Measurement and Data Interpretation

Why does carbon monoxide poisoning cause falsely high pulse oximetry readings?

Answer

Carboxyhaemoglobin absorbs light at a similar wavelength to oxyhaemoglobin, so the pulse oximeter interprets it as oxygenated haemoglobin, overestimating true oxygen saturation.

#378 · Physics Clinical Measurement and Data Interpretation

What effect does methaemoglobinaemia have on pulse oximetry readings?

Answer

Methaemoglobin absorbs light at both wavelengths similarly, driving the reading towards approximately 85 percent regardless of true oxygen saturation.

#379 · Physics Clinical Measurement and Data Interpretation

Name three common sources of error in pulse oximetry besides abnormal haemoglobins.

Answer

Poor peripheral perfusion, motion artefact, nail varnish or skin pigmentation, ambient light interference, and dysrhythmias affecting the pulsatile signal can all cause inaccurate readings.

#380 · Physics Clinical Measurement and Data Interpretation

Why can pulse oximetry fail to detect hypoventilation in a patient receiving supplemental oxygen?

Answer

Supplemental oxygen maintains saturation despite rising carbon dioxide and reduced ventilation, since oxygen reserve delays desaturation, masking early hypoventilation.

#381 · Physics Clinical Measurement and Data Interpretation

Describe the four phases of a normal capnography waveform.

Answer

Phase I is the inspiratory baseline (dead space gas, no carbon dioxide), Phase II is a rapid upstroke as alveolar gas mixes in, Phase III is the alveolar plateau, and Phase IV (downstroke) marks the start of inspiration.

#382 · Physics Clinical Measurement and Data Interpretation

What capnography finding suggests oesophageal intubation?

Answer

A flat trace with absent or minimal, rapidly decaying carbon dioxide readings across a few breaths indicates the tube is not in the trachea.

#383 · Physics Clinical Measurement and Data Interpretation

What capnography pattern indicates rebreathing of carbon dioxide?

Answer

A raised baseline that fails to return to zero during inspiration indicates rebreathing, often from exhausted soda lime or a faulty unidirectional valve in a circle system.

#384 · Physics Clinical Measurement and Data Interpretation

What does a capnography trace with a slowly rising, sloped upstroke and plateau (a 'shark fin' pattern) suggest?

Answer

It suggests bronchospasm or obstructive airway disease, caused by uneven, delayed emptying of alveoli with differing time constants.

#385 · Physics Clinical Measurement and Data Interpretation

What waveform features characterise a normal arterial blood pressure trace, and what does a normal central venous pressure trace show?

Answer

The arterial trace shows a rapid systolic upstroke, a dicrotic notch marking aortic valve closure, and a diastolic runoff, while a normal CVP trace shows a, c and v waves corresponding to atrial contraction, tricuspid valve bulging, and atrial filling respectively.

#386 · Physics Clinical Measurement and Data Interpretation

What is categorical data?

Answer

Data that places observations into named groups with no inherent numerical value, such as blood group or sex. Categorical data can be further divided into nominal (no order) and ordinal (ordered) types.

#387 · Physics Clinical Measurement and Data Interpretation

What is nominal data?

Answer

A type of categorical data in which categories have no natural order, for example eye colour or blood group.

#388 · Physics Clinical Measurement and Data Interpretation

What is ordinal data?

Answer

Categorical data with a meaningful order but unequal or unknown intervals between categories, such as pain score or ASA grade. Median and mode are appropriate summary measures, not the mean.

#389 · Physics Clinical Measurement and Data Interpretation

What is continuous data?

Answer

Numerical data that can take any value within a range, such as height, weight or blood pressure. It may be further classified as interval (no true zero, for example temperature in degrees Celsius) or ratio (true zero, for example weight in kilograms).

#390 · Physics Clinical Measurement and Data Interpretation

What is discrete data?

Answer

Numerical data that can only take fixed, countable values, such as the number of children or number of previous operations. It cannot take fractional values between the counted units.

#391 · Physics Clinical Measurement and Data Interpretation

Define parametric data.

Answer

Data that follows a known distribution, usually the normal distribution, allowing the use of tests that assume this distribution such as the t-test and ANOVA. Parametric tests are generally more powerful when their assumptions are met.

#392 · Physics Clinical Measurement and Data Interpretation

Define non-parametric data.

Answer

Data that does not follow a known distribution, is ordinal, or has a small sample size, requiring distribution-free statistical tests such as the Mann-Whitney U test. Non-parametric tests make fewer assumptions but are generally less powerful.

#393 · Physics Clinical Measurement and Data Interpretation

How can normality of a data set be assessed?

Answer

By plotting a histogram or Q-Q plot to inspect the shape visually, or by formal tests such as the Shapiro-Wilk or Kolmogorov-Smirnov test. A sample size that is sufficiently large also supports the use of parametric assumptions via the central limit theorem.

#394 · Physics Clinical Measurement and Data Interpretation

Define the mean.

Answer

The arithmetic average of a data set, calculated by summing all values and dividing by the number of observations. It is sensitive to outliers and is the appropriate measure of central tendency for normally distributed data.

#395 · Physics Clinical Measurement and Data Interpretation

Define the median.

Answer

The middle value of a data set when all values are arranged in ascending order. It is not affected by outliers and is the appropriate measure of central tendency for skewed or ordinal data.

#396 · Physics Clinical Measurement and Data Interpretation

Define the mode.

Answer

The most frequently occurring value in a data set. A data set may have one mode (unimodal), two modes (bimodal) or more, and it is the only valid measure of central tendency for nominal data.

#397 · Physics Clinical Measurement and Data Interpretation

What is the range?

Answer

The difference between the highest and lowest values in a data set. It is easy to calculate but is heavily influenced by outliers since it uses only the two extreme values.

#398 · Physics Clinical Measurement and Data Interpretation

What is the interquartile range?

Answer

The difference between the 75th percentile (upper quartile) and the 25th percentile (lower quartile) of a data set, describing the spread of the middle 50% of values. It is less affected by outliers than the full range.

#399 · Physics Clinical Measurement and Data Interpretation

Define standard deviation.

Answer

A measure of the spread of individual data points around the mean, calculated as the square root of the variance. It is only meaningfully interpreted for normally distributed data.

#400 · Physics Clinical Measurement and Data Interpretation

Define variance.

Answer

The average of the squared differences between each data point and the mean, representing the spread of the data. Standard deviation is derived by taking the square root of the variance to return to the original units.

#401 · Physics Clinical Measurement and Data Interpretation

What is the standard error of the mean (SEM)?

Answer

A measure of how precisely the sample mean estimates the true population mean, calculated as the standard deviation divided by the square root of the sample size. Unlike standard deviation, SEM decreases as sample size increases.

#402 · Physics Clinical Measurement and Data Interpretation

How does increasing sample size affect the standard error of the mean?

Answer

The standard error of the mean falls as sample size increases, because it is inversely proportional to the square root of the sample size. This reflects greater precision in estimating the true population mean, not less variability in individual data points.

#403 · Physics Clinical Measurement and Data Interpretation

What is a confidence interval?

Answer

A range of values, calculated from sample data, within which the true population parameter is expected to lie with a stated level of confidence, most commonly 95%. A 95% confidence interval means that if the study were repeated many times, 95% of the calculated intervals would contain the true population value.

#404 · Physics Clinical Measurement and Data Interpretation

How is a 95% confidence interval for the mean calculated?

Answer

It is calculated as the sample mean plus or minus approximately 1.96 multiplied by the standard error of the mean. A narrower confidence interval indicates a more precise estimate of the true population value.

#405 · Physics Clinical Measurement and Data Interpretation

What does it mean if a confidence interval for a difference between two treatments crosses zero?

Answer

It indicates that there is no statistically significant difference between the two treatments, since a difference of zero (no effect) is a plausible value. This is analogous to a non-significant p-value.

#406 · Physics Clinical Measurement and Data Interpretation

Describe the shape of the normal distribution.

Answer

A symmetrical, bell-shaped curve in which the mean, median and mode are all equal and located at the centre. It is defined entirely by its mean and standard deviation.

#407 · Physics Clinical Measurement and Data Interpretation

What proportion of data lies within 1, 2 and 3 standard deviations of the mean in a normal distribution?

Answer

Approximately 68% of data lies within 1 standard deviation of the mean, 95% within 2 standard deviations, and 99.7% within 3 standard deviations. This is known as the empirical or 68-95-99.7 rule.

#408 · Physics Clinical Measurement and Data Interpretation

What is a skewed distribution?

Answer

A distribution in which data are asymmetrically spread, with a longer tail on one side. In a positively (right) skewed distribution the mean is greater than the median, while in a negatively (left) skewed distribution the mean is less than the median.

#409 · Physics Clinical Measurement and Data Interpretation

State the central limit theorem.

Answer

As sample size increases, the distribution of sample means approaches a normal distribution regardless of the shape of the underlying population distribution. This justifies the use of parametric tests on sufficiently large samples even when the original data are not normally distributed.

#410 · Physics Clinical Measurement and Data Interpretation

Define the null hypothesis.

Answer

A statement assuming there is no true difference or association between the groups or variables being studied. Statistical tests are designed to assess the evidence against this hypothesis.

#411 · Physics Clinical Measurement and Data Interpretation

Define the alternative hypothesis.

Answer

A statement proposing that there is a true difference or association between the groups or variables being studied. It is accepted only if the null hypothesis is rejected on the basis of the statistical test result.

#412 · Physics Clinical Measurement and Data Interpretation

What is a p-value?

Answer

The probability of obtaining a result as extreme as, or more extreme than, the one observed, if the null hypothesis were true. A p-value below the chosen significance threshold, usually 0.05, leads to rejection of the null hypothesis.

#413 · Physics Clinical Measurement and Data Interpretation

Define a Type I error.

Answer

The incorrect rejection of a true null hypothesis, concluding that a difference exists when it does not. The probability of a Type I error is denoted alpha and is conventionally set at 0.05.

#414 · Physics Clinical Measurement and Data Interpretation

Define a Type II error.

Answer

The incorrect failure to reject a false null hypothesis, concluding that no difference exists when one truly does. The probability of a Type II error is denoted beta and is commonly set at 0.2 (20%).

#415 · Physics Clinical Measurement and Data Interpretation

Define statistical power.

Answer

The probability that a study will correctly detect a true difference or effect when one genuinely exists, calculated as 1 minus beta. A power of 80% is a commonly accepted minimum for clinical trials.

#416 · Physics Clinical Measurement and Data Interpretation

Name three factors that increase the statistical power of a study.

Answer

Increasing the sample size, increasing the size of the true effect being measured, and reducing variability within the data all increase statistical power. Using a less stringent significance level also increases power but raises the risk of a Type I error.

#417 · Physics Clinical Measurement and Data Interpretation

What is a one-tailed statistical test?

Answer

A test that assesses whether a result is significantly different from the null hypothesis in one specified direction only, for example whether a drug reduces blood pressure. It should only be used when a difference in the opposite direction is genuinely implausible or clinically irrelevant.

#418 · Physics Clinical Measurement and Data Interpretation

What is a two-tailed statistical test?

Answer

A test that assesses whether a result is significantly different from the null hypothesis in either direction, without specifying which. It is the more conservative and generally preferred approach in clinical research.

#419 · Physics Clinical Measurement and Data Interpretation

Which statistical test compares the means of two independent groups of normally distributed data?

Answer

The unpaired (independent samples) Student's t-test. It requires the data in each group to be continuous, approximately normally distributed and to have similar variances.

#420 · Physics Clinical Measurement and Data Interpretation

Which statistical test compares two sets of normally distributed measurements taken from the same subjects?

Answer

The paired Student's t-test, which analyses the differences between paired observations, such as blood pressure before and after an intervention in the same patients. It is more powerful than an unpaired test when data are genuinely paired.

#421 · Physics Clinical Measurement and Data Interpretation

Which test compares two independent groups of non-parametric or ordinal data?

Answer

The Mann-Whitney U test, a non-parametric equivalent of the unpaired t-test. It compares the ranks of values between the two groups rather than the raw data itself.

#422 · Physics Clinical Measurement and Data Interpretation

Which test compares two related sets of non-parametric or ordinal data?

Answer

The Wilcoxon signed-rank test, a non-parametric equivalent of the paired t-test. It is used when the same subjects are measured twice under different conditions and the data are not normally distributed.

#423 · Physics Clinical Measurement and Data Interpretation

Which test is used to compare categorical data between two or more independent groups?

Answer

The chi-squared test, which compares observed frequencies against those expected under the null hypothesis. Fisher's exact test should be used instead when expected cell frequencies are small, typically less than 5.

#424 · Physics Clinical Measurement and Data Interpretation

Which test compares the means of three or more independent groups of normally distributed data?

Answer

One-way analysis of variance (ANOVA), which tests whether at least one group mean differs significantly from the others. A significant result requires post-hoc testing to identify exactly which groups differ.

#425 · Physics Clinical Measurement and Data Interpretation

Which test is the non-parametric equivalent of one-way ANOVA?

Answer

The Kruskal-Wallis test, used to compare three or more independent groups of non-parametric or ordinal data. It compares the ranks of values across all groups rather than the raw values.

#426 · Physics Clinical Measurement and Data Interpretation

Which statistical test measures the strength and direction of a linear association between two continuous variables?

Answer

Pearson's correlation coefficient, denoted r , ranging from -1 (perfect negative correlation) to +1 (perfect positive correlation), with 0 indicating no linear correlation. Spearman's rank correlation is used instead for non-parametric or ordinal data.

#427 · Physics Clinical Measurement and Data Interpretation

Define sensitivity of a diagnostic test.

Answer

The proportion of patients who truly have the disease who are correctly identified as positive by the test, calculated as true positives divided by (true positives plus false negatives). A highly sensitive test is useful for ruling out disease when negative.

#428 · Physics Clinical Measurement and Data Interpretation

Define specificity of a diagnostic test.

Answer

The proportion of patients who truly do not have the disease who are correctly identified as negative by the test, calculated as true negatives divided by (true negatives plus false positives). A highly specific test is useful for ruling in disease when positive.

#429 · Physics Clinical Measurement and Data Interpretation

Define positive predictive value (PPV).

Answer

The proportion of patients with a positive test result who truly have the disease, calculated as true positives divided by (true positives plus false positives). Unlike sensitivity and specificity, PPV varies with the prevalence of disease in the population tested.

#430 · Physics Clinical Measurement and Data Interpretation

Define negative predictive value (NPV).

Answer

The proportion of patients with a negative test result who truly do not have the disease, calculated as true negatives divided by (true negatives plus false negatives). Like PPV, NPV changes with disease prevalence.

#431 · Physics Clinical Measurement and Data Interpretation

How does disease prevalence affect positive predictive value?

Answer

As disease prevalence falls, the positive predictive value of a test falls even if sensitivity and specificity remain unchanged, because a greater proportion of positive results become false positives. This is why screening tests in low-prevalence populations generate many false positives.

#432 · Physics Clinical Measurement and Data Interpretation

How does disease prevalence affect negative predictive value?

Answer

As disease prevalence rises, the negative predictive value of a test falls even if sensitivity and specificity remain unchanged, because a greater proportion of negative results become false negatives. Sensitivity and specificity themselves are properties of the test and do not change with prevalence.

#433 · Physics Clinical Measurement and Data Interpretation

What is a receiver operating characteristic (ROC) curve used for?

Answer

It plots sensitivity against 1 minus specificity across a range of diagnostic thresholds, allowing comparison of test performance and selection of the optimal cut-off value. The area under the curve reflects overall test accuracy, with 1.0 representing a perfect test and 0.5 representing no better than chance.

#434 · Physics Clinical Measurement and Data Interpretation

Describe the design of a randomised controlled trial (RCT).

Answer

Participants are randomly allocated to receive either the intervention or a comparator (placebo or standard treatment), and outcomes are compared prospectively between groups. Randomisation minimises confounding and selection bias, making the RCT the gold-standard design for testing causation.

#435 · Physics Clinical Measurement and Data Interpretation

Describe a cohort study.

Answer

A group of individuals with a shared exposure characteristic is followed forward in time (prospectively) or identified retrospectively to compare the incidence of an outcome against an unexposed group. It can estimate relative risk directly but is more prone to confounding than an RCT.

#436 · Physics Clinical Measurement and Data Interpretation

Describe a case-control study.

Answer

Patients who already have a particular outcome (cases) are compared retrospectively with those who do not (controls) to identify differences in past exposures. It is well suited to studying rare diseases but can only estimate an odds ratio, not relative risk directly, and is prone to recall bias.

#437 · Physics Clinical Measurement and Data Interpretation

Define selection bias.

Answer

A systematic error arising when the individuals included in a study are not representative of the population intended to be studied, distorting the true association between exposure and outcome. It can occur through non-random sampling or differential loss to follow-up.

#438 · Physics Clinical Measurement and Data Interpretation

Define recall bias.

Answer

A form of information bias in which participants with a particular outcome, such as a disease, remember and report past exposures differently or more thoroughly than those without the outcome. It is a particular problem in retrospective study designs such as case-control studies.

#439 · Physics Clinical Measurement and Data Interpretation

Define confounding.

Answer

A situation in which the observed association between an exposure and an outcome is distorted by a third factor that is independently related to both the exposure and the outcome. Randomisation in an RCT is the most effective method of controlling for both known and unknown confounders.

#440 · Physics Clinical Measurement and Data Interpretation

What are the stages of the clinical audit cycle?

Answer

Setting a standard, measuring current practice against that standard, implementing change to address any shortfall, and then re-auditing to confirm the change produced the intended improvement, closing the loop.

#441 · Pharmacology

Which cytochrome enzyme metabolises fentanyl, and what interaction risk follows?

Answer

Fentanyl is metabolised by CYP3A4.
Co-administration with potent CYP3A4 inhibitors such as clarithromycin, ritonavir or itraconazole can markedly reduce clearance and prolong opioid effect.

#442 · Pharmacology

Why does erythromycin prolong the action of midazolam?

Answer

Erythromycin inhibits hepatic CYP3A4, the enzyme responsible for midazolam oxidation, reducing its clearance and producing exaggerated or prolonged sedation.

#443 · Pharmacology

What is the mechanism of the interaction between suxamethonium and anticholinesterases such as neostigmine?

Answer

Neostigmine inhibits plasma cholinesterase as well as acetylcholinesterase, so it slows suxamethonium hydrolysis and can prolong the block rather than reverse it.

#444 · Pharmacology

What type of pharmacokinetic model is the Marsh model for propofol TCI, and how does it scale its parameters?

Answer

The Marsh model is a fixed-parameter, weight-based three-compartment model for propofol TCI; volumes and clearances scale linearly with total body weight and do not adjust for age, so it can overshoot in the elderly.

#445 · Pharmacology

How does the Schnider model differ from the Marsh model for propofol TCI?

Answer

The Schnider model incorporates age, height, weight and lean body mass into its pharmacokinetic parameters and targets effect-site rather than plasma concentration, giving more accurate dosing in elderly or obese patients.

#446 · Pharmacology

What population does the Paedfusor model apply to for propofol TCI?

Answer

The Paedfusor model is validated for children aged 1 to 16 years weighing 5 to 61 kg, using a modified three-compartment structure with weight-based parameters distinct from Marsh.

#447 · Pharmacology

What is the Minto model used for in TIVA?

Answer

The Minto model is the pharmacokinetic-pharmacodynamic model for remifentanyl TCI, incorporating age and lean body mass to predict plasma and effect-site concentrations.

#448 · Pharmacology

Why is lean body mass rather than total body weight used in some TCI models?

Answer

In obese patients, using total body weight in models like Schnider or Minto can cause substantial overdosing because fat contributes little to the central volume of distribution, so lean body mass gives a more accurate effect-site target.

#449 · Pharmacology

What propofol plasma target typically achieves loss of consciousness using the Marsh model?

Answer

A plasma target of around 4 to 6 micrograms per millilitre typically produces loss of consciousness with the Marsh model, though clinical titration against response is essential.

#450 · Pharmacology

How should morphine dosing be adjusted in significant renal impairment?

Answer

Morphine's active metabolite morphine-6-glucuronide accumulates in renal failure and can cause prolonged sedation and respiratory depression, so the dose should be reduced and the interval extended, or an alternative such as fentanyl used.

#451 · Pharmacology

Why is fentanyl often preferred over morphine in patients with renal failure?

Answer

Fentanyl has no active renally-excreted metabolites, unlike morphine, so its pharmacokinetics are less affected by reduced renal clearance.

#452 · Pharmacology

How does hepatic cirrhosis affect rocuronium dosing?

Answer

Rocuronium is largely eliminated by hepatic and biliary excretion, so cirrhosis increases its volume of distribution and prolongs duration of action, requiring reduced dosing or careful neuromuscular monitoring.

#453 · Pharmacology

Which neuromuscular blocking drug is preferred in significant hepatic and renal failure, and why?

Answer

Atracurium (or cisatracurium) is preferred because it undergoes Hofmann elimination and ester hydrolysis, processes independent of hepatic or renal function.

#454 · Pharmacology

How should induction doses of propofol be adjusted in elderly patients?

Answer

Propofol induction dose should be reduced by around 20 to 50% in the elderly because of reduced central volume of distribution, decreased clearance and increased brain sensitivity.

#455 · Pharmacology

Why do neonates require higher weight-adjusted doses of some neuromuscular blockers despite immature neuromuscular junctions?

Answer

Neonates have a larger extracellular fluid volume and hence a larger volume of distribution per kilogram, which can offset their junctional immaturity and altered sensitivity, so dosing must be individualised rather than simply scaled down.

#456 · Pharmacology

Which local anaesthetic and maximum dose is commonly used for spinal anaesthesia in obstetrics, and why is total dose reduced in pregnancy?

Answer

Hyperbaric bupivacaine is commonly used; pregnant women require lower doses because epidural venous engorgement reduces cerebrospinal fluid volume and progesterone increases neural sensitivity to local anaesthetic.

#457 · Pharmacology

Why is suxamethonium dose per kilogram sometimes increased in pregnancy despite reduced plasma cholinesterase activity?

Answer

Pregnancy increases volume of distribution, so despite lower plasma cholinesterase activity a standard or slightly higher weight-based dose is still used to ensure adequate rapid sequence intubation conditions.

#458 · Pharmacology

What is the mechanism of action of ondansetron as an antiemetic?

Answer

Ondansetron is a selective 5-HT₃ receptor antagonist, blocking serotonin receptors both peripherally on vagal afferents in the gut and centrally in the chemoreceptor trigger zone.

#459 · Pharmacology

What is the mechanism of action of cyclizine as an antiemetic?

Answer

Cyclizine is an antihistamine (H1 receptor antagonist) with additional antimuscarinic activity, acting mainly on the vomiting centre and vestibular pathways.

#460 · Pharmacology

What receptor does dexamethasone act on to reduce postoperative nausea and vomiting?

Answer

The precise antiemetic mechanism of dexamethasone is not fully defined, but it is thought to act via glucocorticoid receptors reducing prostaglandin and central serotonin release, alongside anti-inflammatory effects.

#461 · Pharmacology

What is the mechanism of action of droperidol as an antiemetic, and its key safety concern?

Answer

Droperidol is a dopamine D2 receptor antagonist acting on the chemoreceptor trigger zone; its key concern is dose-dependent QT interval prolongation and risk of torsades de pointes.

#462 · Pharmacology

What is the mechanism of action of aprepitant in preventing postoperative nausea and vomiting?

Answer

Aprepitant is a neurokinin-1 (NK1) receptor antagonist that blocks substance P binding in the chemoreceptor trigger zone and vomiting centre.

#463 · Pharmacology

How long before a neuraxial block should prophylactic-dose low molecular weight heparin be stopped?

Answer

Prophylactic-dose LMWH should be stopped at least 12 hours before neuraxial block or catheter removal, and treatment-dose LMWH at least 24 hours before.

#464 · Pharmacology

How long after a neuraxial block or catheter removal can prophylactic LMWH be restarted?

Answer

Prophylactic LMWH can typically be restarted at least 4 hours after neuraxial block or catheter removal, provided there is no evidence of bleeding.

#465 · Pharmacology

What INR threshold is generally considered safe for neuraxial block in a patient on warfarin?

Answer

An INR of 1.4 or less is generally regarded as an acceptable threshold for safe neuraxial block, with warfarin usually stopped around 5 days beforehand.

#466 · Pharmacology

How long before neuraxial block should apixaban or rivaroxaban be stopped for a normal renal function patient?

Answer

Apixaban and rivaroxaban should generally be stopped at least 3 days (72 hours) before neuraxial block in patients with normal renal function, reflecting AAGBI/regional anaesthesia guidance.

#467 · Pharmacology

How long after neuraxial catheter removal should a therapeutic dose DOAC be restarted?

Answer

A therapeutic dose DOAC should generally not be restarted until at least 24 hours after catheter removal, longer than the roughly 6 hour interval used for restarting a prophylactic dose, reflecting the higher bleeding risk of a full anticoagulant dose so soon after instrumentation.

#468 · Pharmacology

How long before neuraxial block should clopidogrel be stopped?

Answer

Clopidogrel should be stopped for at least 7 days before neuraxial block because of its irreversible antiplatelet effect on platelet P2Y₁₂ receptors.

#469 · Pharmacology

Does aspirin monotherapy need to be stopped before neuraxial block?

Answer

No, aspirin monotherapy at standard antiplatelet doses is not considered a contraindication to neuraxial block and does not need to be stopped.

#470 · Pharmacology

What is the mechanism by which sugammadex reverses rocuronium, and why is it ineffective against benzyloquinolinium blockers?

Answer

Sugammadex is a modified gamma-cyclodextrin that encapsulates rocuronium (and vecuronium) molecules in a tight host-guest complex, preventing them from acting at the neuromuscular junction; it does not effectively encapsulate benzyloquinoliniums such as atracurium because their shape does not fit the cyclodextrin cavity.

#471 · Physiology Anatomy and Biochemistry

What dermatome level corresponds to the nipple line?

Answer

The nipple line corresponds to dermatome T4. This landmark is commonly used to assess the upper level of a spinal or epidural block for caesarean section.

#472 · Physiology Anatomy and Biochemistry

What dermatome level corresponds to the xiphisternum?

Answer

The xiphisternum corresponds to dermatome T6. Blockade to this level is often the target for upper abdominal surgery under neuraxial anaesthesia.

#473 · Physiology Anatomy and Biochemistry

What dermatome level corresponds to the umbilicus?

Answer

The umbilicus corresponds to dermatome T10. This level is a common target block height for lower abdominal procedures and labour analgesia.

#474 · Physiology Anatomy and Biochemistry

Which branch of the brachial plexus supplies the deltoid and teres minor muscles?

Answer

The axillary nerve, arising from the posterior cord, supplies the deltoid and teres minor muscles and provides sensation over the regimental badge area of the upper lateral arm.

#475 · Physiology Anatomy and Biochemistry

Which brachial plexus branch is responsible for elbow flexion via biceps brachii?

Answer

The musculocutaneous nerve, arising from the lateral cord, supplies biceps brachii and brachialis, providing the main power for elbow flexion, and continues as the lateral cutaneous nerve of the forearm.

#476 · Physiology Anatomy and Biochemistry

Which nerve of the brachial plexus is most commonly injured by prolonged pressure on the medial epicondyle of the humerus?

Answer

The ulnar nerve, arising from the medial cord, runs behind the medial epicondyle and is vulnerable to compression injury during prolonged arm positioning under anaesthesia.

#477 · Physiology Anatomy and Biochemistry

What is the P50 of haemoglobin and what does it represent?

Answer

P50 is the partial pressure of oxygen at which haemoglobin is 50% saturated, normally about 3.5 kPa (26.6 mmHg). A lower P50 indicates a left-shifted curve with increased oxygen affinity.

#478 · Physiology Anatomy and Biochemistry

Name three factors that shift the oxyhaemoglobin dissociation curve to the right.

Answer

A rise in temperature, a rise in 2,3-diphosphoglycerate, and a fall in pH (increased hydrogen ion concentration, the Bohr effect) all shift the curve to the right, reducing haemoglobin oxygen affinity.

#479 · Physiology Anatomy and Biochemistry

What are the normal ranges for intracranial pressure and cerebral perfusion pressure?

Answer

Normal intracranial pressure is 7 to 15 mmHg in a supine adult, and normal cerebral perfusion pressure is 60 to 80 mmHg, calculated as mean arterial pressure minus intracranial pressure.

#480 · Physiology Anatomy and Biochemistry

Over what range of cerebral perfusion pressure is cerebral blood flow autoregulated?

Answer

Cerebral blood flow is autoregulated between a cerebral perfusion pressure of approximately 60 and 150 mmHg, remaining relatively constant across this range through myogenic adjustment of vascular tone.

#481 · Physiology Anatomy and Biochemistry

What are the normal reference ranges for arterial pH, PaCO₂ and PaO₂ on an arterial blood gas?

Answer

Normal arterial pH is 7.35 to 7.45, PaCO₂ is 4.6 to 6.0 kPa, and PaO₂ is 10.6 to 13.3 kPa when breathing room air at sea level.

#482 · Physiology Anatomy and Biochemistry

What is the normal range for standard base excess on an arterial blood gas?

Answer

The normal range for standard base excess is minus 2 to plus 2 mmol per litre. A value more negative than minus 2 indicates a metabolic acidosis and more positive than plus 2 indicates a metabolic alkalosis.

#483 · Physiology Anatomy and Biochemistry

What is the normal anion gap and how is it calculated?

Answer

The anion gap is calculated as (sodium plus potassium) minus (chloride plus bicarbonate), with a normal range of approximately 12 to 20 mmol/L when potassium is included, higher than the 8 to 16 mmol/L range used when potassium is excluded from the calculation.

#484 · Physiology Anatomy and Biochemistry

Which spinal cord level marks the normal termination of the adult spinal cord (conus medullaris)?

Answer

The adult spinal cord normally terminates at the level of the L1/L2 vertebral interspace, which is why lumbar puncture and spinal anaesthesia are performed below this level to avoid direct cord injury.

#485 · Physiology Anatomy and Biochemistry

At what vertebral level does the trachea bifurcate into the two main bronchi?

Answer

The trachea bifurcates at the carina, at the level of the sternal angle anteriorly, corresponding to the T4/T5 vertebral level posteriorly.

#486 · Physiology Anatomy and Biochemistry

What is the normal adult functional residual capacity and which two lung volumes make it up?

Answer

Functional residual capacity is normally about 2.5 to 3.0 litres in an adult and is made up of the expiratory reserve volume plus the residual volume.

#487 · Physiology Anatomy and Biochemistry

What proportion of total body oxygen consumption does the brain normally account for at rest?

Answer

The brain accounts for approximately 20% of total body oxygen consumption at rest despite representing only about 2% of total body weight.

#488 · Physiology Anatomy and Biochemistry

What is the normal glomerular filtration rate in a healthy adult?

Answer

Normal glomerular filtration rate in a healthy adult is approximately 100 to 120 mL per minute per 1.73 square metres of body surface area.

#489 · Physiology Anatomy and Biochemistry

What is the normal osmolar gap and what does a raised gap suggest?

Answer

The normal osmolar gap, the difference between measured and calculated plasma osmolality, is less than 10 mOsm per kg. A raised gap suggests the presence of an unmeasured osmotically active substance such as ethanol, methanol or ethylene glycol.

#490 · Physiology Anatomy and Biochemistry

Which coagulation factors are vitamin K dependent?

Answer

Factors II, VII, IX and X, along with proteins C and S, are vitamin K dependent and are synthesised in the liver by gamma-carboxylation.

#491 · Physics Clinical Measurement and Data Interpretation

What is the critical temperature of nitrous oxide?

Answer

The critical temperature of nitrous oxide is 36.5 degrees Celsius, the temperature above which it cannot exist as a liquid regardless of the pressure applied.

#492 · Physics Clinical Measurement and Data Interpretation

What colour and size are standard UK oxygen cylinders, and what is a full size E cylinder pressure?

Answer

UK oxygen cylinders are white (shoulder and body under the current standard), and a full size E cylinder is filled to approximately 137 bar, holding about 680 litres of gas.

#493 · Physics Clinical Measurement and Data Interpretation

What colour is a UK Entonox cylinder and what does it contain?

Answer

Entonox cylinders have a white body with a blue and white quartered shoulder, containing a 50:50 mixture by volume of oxygen and nitrous oxide.

#494 · Physics Clinical Measurement and Data Interpretation

What colour is a UK medical air cylinder?

Answer

UK medical air cylinders have a white body with black and white quartered shoulders under the current ISO 32 standard.

#495 · Physics Clinical Measurement and Data Interpretation

At what Reynolds number does gas flow typically transition from laminar to turbulent?

Answer

Flow typically becomes turbulent above a Reynolds number of approximately 4000; below approximately 2000 it remains laminar, with an unpredictable transitional zone in between.

#496 · Physics Clinical Measurement and Data Interpretation

What is the normal end-tidal carbon dioxide value on capnography?

Answer

Normal end-tidal carbon dioxide is approximately 4.6 to 6.0 kPa (35 to 45 mmHg), closely reflecting arterial PaCO₂ in a patient with normal ventilation-perfusion matching.

#497 · Physics Clinical Measurement and Data Interpretation

What p-value is conventionally used as the threshold for statistical significance, and what does it mean?

Answer

A p-value of less than 0.05 is the conventional threshold for statistical significance, meaning there is less than a 5% probability that the observed result occurred by chance alone if the null hypothesis were true.

#498 · Physics Clinical Measurement and Data Interpretation

What does a type II statistical error represent?

Answer

A type II error is the failure to reject a null hypothesis that is actually false, in effect a false negative result, and its probability is denoted beta. Statistical power is calculated as 1 minus beta.

#499 · Physics Clinical Measurement and Data Interpretation

During the pre-use anaesthetic machine check, what pressure is used for the leak test on a circle breathing system?

Answer

The breathing system is pressurised to approximately 30 cmH₂O with the adjustable pressure limiting valve closed and the patient end occluded, and the pressure should be held for at least 10 seconds without falling to confirm no leak.

#500 · Physics Clinical Measurement and Data Interpretation

What is the purpose of the oxygen failure warning device (Ritchie whistle) on an anaesthetic machine?

Answer

The oxygen failure warning device sounds an audible alarm and cuts off the supply of other gases when the oxygen supply pressure falls below approximately 200 kPa, preventing delivery of a hypoxic gas mixture.